

ROADS INVESTMENT PLAN 2016 - 2020 NEEDS ANALYSIS FOR THE ROAD NETWORK



Uganda National Roads Authority

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Executive Summary

The principle objective of this strategic level analysis was to support the preparation of an Investment Plan for the UNRA road network for a five year period spanning 2016 to 2020. The specific objectives were to:

- Show the impact of the current funding level on the future road network condition, taking into account current committed projects;
- Show the impact of a higher funding scenario on future road conditions;
- Assess the impact of the funding scenarios on maintenance backlog, road user costs, asset value; and
- Prepare a five year optimised Preventive Maintenance and Rehabilitation Plan for the current funding level.

This report contains the results of the analysis and is structured in two components, consisting of a ‘situational analysis’ and a ‘needs analysis’.

Situational analysis

The total length of the national road network investigated in this report is based on the length of road with complete visual condition assessment information in 2015. In total 16,203 km of road is investigated of which 2,881 km are paved roads and 13,322 km are unpaved roads. This is 79% of the total road network, being 69% of the paved roads and 82% of the unpaved road network.

Traffic volumes on the national network are very high, approximately 75% of the paved road network carries more than 5,000 vehicles per day and 95% of the unpaved roads carry more than 500 vehicles per day. The high traffic volumes on the unpaved network imply a high need for upgrading to paved standards.

According to the most recent condition assessment data of 2015, the overall condition of the paved roads can be classified as “good”. The average condition improved since 2009 / 2011. Only 2% of the network consists of poor and very poor roads, indicating a low rehabilitation

backlog. The overall condition of the unpaved roads can be classified as “fair”. The average condition has remained the same since 2009 / 2011. The 25% of poor and very poor roads have high investment needs and these roads are functioning with little or no gravel wearing course material. The current average gravel thickness of the unpaved roads is 65mm.

Needs analysis

This analysis determined the impact of the current funding levels considering the following intervention treatments: reseal, rehabilitation, re-gravelling and upgrading of unpaved roads to paved standards.

The analysis derived at the following outcomes:

“

- **The current funding level of US\$ 843 per annum is sufficient to maintain the road network, whether paved or unpaved, at current conditions, and will increase the performance levels over the analysis period.**
- **The optimal allocation of the current funding to paved roads is 18% (US\$ 154 million per annum); and these roads will be maintained in the current condition category of “good” over the next 10 years.**
- **The remaining 62% (US\$ 689 million per annum) of the current funding will increase the gravel thickness of unpaved roads to 80mm over the 10 year analysis period. US\$40 million per annum is allocated towards re-graveling, and US\$649 million per annum towards upgrade to paved standards. ”**

While the network can be successfully maintained with the current funding, the excessive need for upgrading of unpaved roads to paved standards is not fully met. Therefore, when comparing the analysis allocation between the current received and a higher funding scenario (‘required funding’ of US\$1,439 million per annum), the main difference in the optimal

funding allocation is the increased funding allocation toward unpaved roads, and specifically toward upgrading to paved standards.

Recommendation

It is recommended that the funding level of US\$43 per annum at least be continued for the next 10 years, while additional funding is sourced to address the high needs of upgrading unpaved roads to paved standards.

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1 Introduction

This document summarises the life cycle cost analysis (LCCA) conducted for the road network of the Uganda National Roads Authority (UNRA) for a ten year analysis period. The purpose of the analysis was to prepare a five year investment plan for the road network. The analysis for this report was done using the dTIMS software customised with the World Bank's Highway Development and Management (HDM-4) pavement performance prediction algorithms. The optimisation system (dTIMS) predicts future pavement performance and consequences of alternative funding and policy scenarios. Multi-year preventive maintenance, rehabilitation and upgrading programs are then generated for given funding scenarios. These programs can be used as basis for compiling the annual works programs.

The investment needs described in this analysis and reports are relevant only for preventive maintenance, rehabilitation and upgrading to paved standards. Needs for interventions such as routine maintenance, capacity and safety improvements, structures, etc are not included in the funding needs determined by this analysis.

2 Report Outline

The report is structured in two major components, each consisting of a number of chapters. The components are:

- Situational analysis (chapters 3 to 6)
- Needs analysis (chapters 7 to 13)

Table 2-1 provides the report outline with the contents per chapter.

TABLE 2-1: REPORT OUTLINE	
CHAPTER	CONTENT
3	The road network under UNRA's jurisdiction is described in terms of the pavement type as well as the roads for which data is available that will be used for the analysis.
4	The extent of traffic usage of the road network is presented.
5	The current condition of the paved road network, based on visual assessment data, is presented and compared with historic data. Measured roughness data is also included.
6	The current condition of the unpaved road network, based on visual assessment data, is presented and compared with historic data.
7	Input parameters and background to the life cycle costs analysis (LCCA) are discussed (treatments, treatment unit costs, budgets, performance prediction models, etc).
8 - 10	These chapters investigate the Technical Needs scenario to determine the funding level required to remove the current backlog in maintenance and rehabilitation needs. Other funding scenarios are investigated in terms of the optimisation of funds towards maintenance and rehabilitation of paved and unpaved roads.
11	The different funding scenarios are compared in terms of paved road condition, gravel thickness and the differences in fund allocations are assessed.
12	Vehicle operating costs and excess user costs for different funding scenarios are assessed.

14	The current asset value, change in asset value and maximum asset value are investigated.
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3 UNRA Road Network

Summary:

- **UNRA has 20,544 km of road under its jurisdiction.**
- **20% (4,157 km) is paved, and 80% (16,388 km) unpaved.**
- **2,881 km (69%) of paved roads was analysed in this report.**
- **13,322 km (82%) unpaved roads was analysed.**

According to the UNRA RMS (Road Management System), the total length of the national road network is 20,544 km. Of these 4,157 km are paved roads and 16,388 km unpaved roads.

Figure 3-1 shows the split between paved (20%) and unpaved (80%) roads.

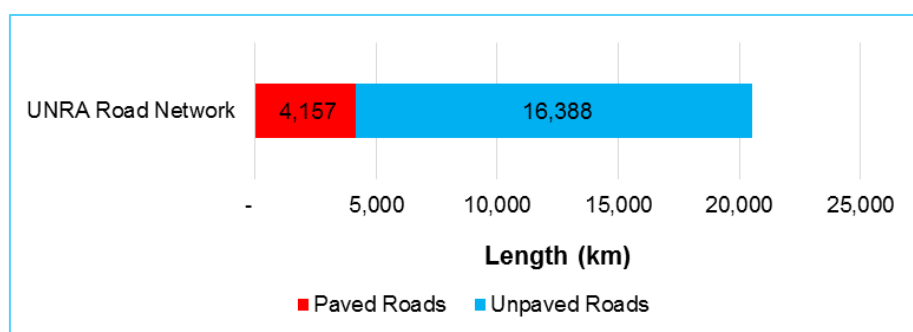


Figure 3-1: Paved and unpaved road lengths of UNRA road network (2015)

The total length of the national road network investigated in this analysis is based on the length of road with complete visual assessment information in 2015. In total 16,203 km of road is investigated of which 2,881 km are paved roads and 13,322 km are unpaved roads. This is 79% of the total road network, being 69% of the paved roads and 82% of the unpaved road network. All condition data was collected internally by UNRA.

Note that although it is preferable that data for the full road network be available for any analysis, this is influenced by roads being under construction at the time of the surveys, and changes or corrections to the roads register.

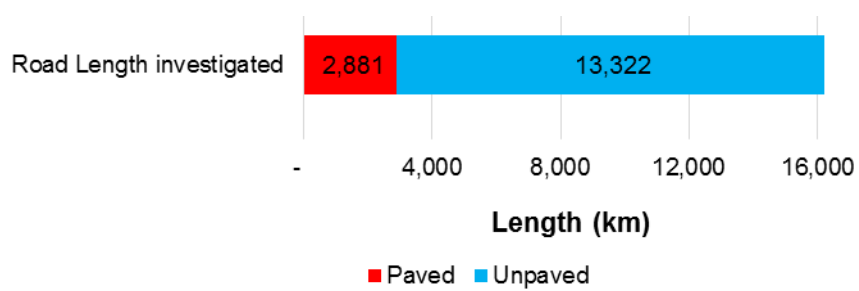


Figure 3-2: Paved and unpaved road lengths investigated (km), 2015

4 Traffic Information

Summary:

- *Approximately, 17,821 million vehicle-kilometres are travelled annually on the UNRA road network.*
- *Approximately 6,697 million vehicle-kilometres are travelled annually on paved roads and approximately 11,125 million vehicle-kilometres annually on unpaved roads.*
- *Slightly more than half the paved network carries less than 5,000 vehicles per day, while another quarter carries 5000 to 10,000 vehicles per day. The last quarter carries more than 10,000 vehicles per day.*
- *Overall, 95% of unpaved roads carry more than 500 vehicles per day.*

Traffic information was obtained from the Traffic Information System (TIS) of the UNRA RMS. The most recent update of the TIS was due to addition of traffic count data collected by UNRA staff in the first quarter of 2016 at approximately 80 manual counting stations. This data was used in the TIS to allocate traffic counts to every road link in the network based on the mother / daughter approach followed in the TIS.

Traffic information is expressed as the Annual Average Daily Traffic (AADT), travelling on each road link of the road network. Furthermore, the product of the AADT on a road link and its length represents the daily vehicle-kilometres travelled on that link in one day. Summarised for the road network, they represent the total distance travelled by vehicles on the road network on a daily or yearly basis. Approximately, 17,821 million vehicle-kilometres are travelled each year on the national road network. Approximately 6,697 million vehicle-kilometres are travelled annually on paved roads and approximately 11,125 million vehicle-kilometres annually on unpaved roads.

Figure 4-1 shows that although the investigated paved road network makes up less than 20% of the total road network, it carries 38% of the network's traffic. A focus on the maintenance and rehabilitation of paved roads will therefore influence more road users per kilometre than unpaved roads.

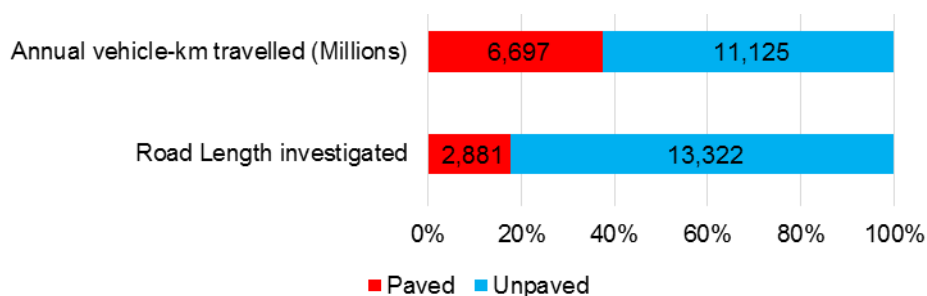


Figure 4-1: Network length investigated versus vehicle-kilometres travelled, 2016

4.1 Traffic per road type

The distribution of traffic volumes for paved and unpaved roads is given in Figure 4-2 and Figure 4-3 respectively. These figures show the portions of network carrying low to high categories of traffic. Slightly more than half the paved network carries less than 5,000 vehicles per day, while another quarter carries 5,000 to 10,000 vehicles per day. The last quarter carries more than 10,000 vehicles per day. For unpaved roads, most of the network (83%) carries between 1,000 and 5,000 vehicles per day. Approximately 11% carries less than 1,000 vehicles per day, and 6% carries more than 5,000 vehicles per day. Overall, 95% of unpaved roads carry more than 500 vehicles per day.

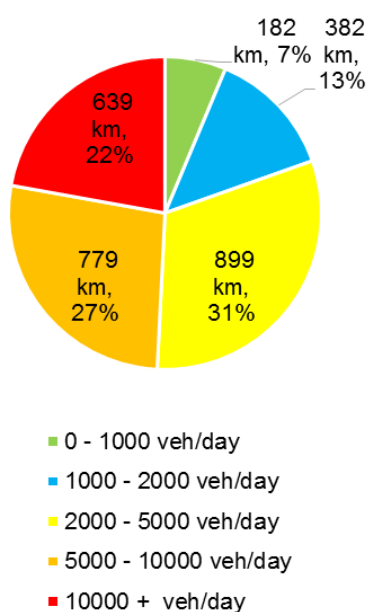


Figure 4-2: Traffic distribution on paved roads, 2016

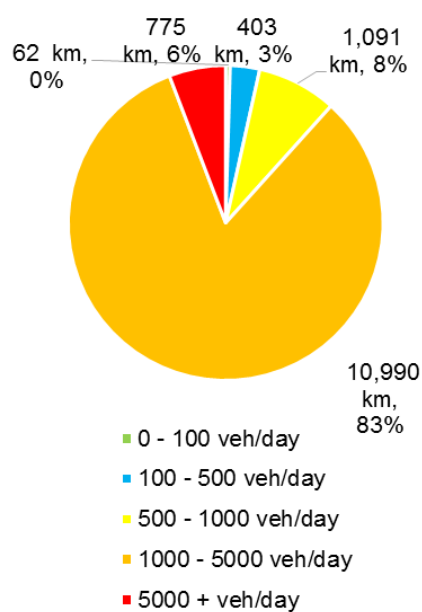


Figure 4-3: Traffic distribution on unpaved roads, 2016

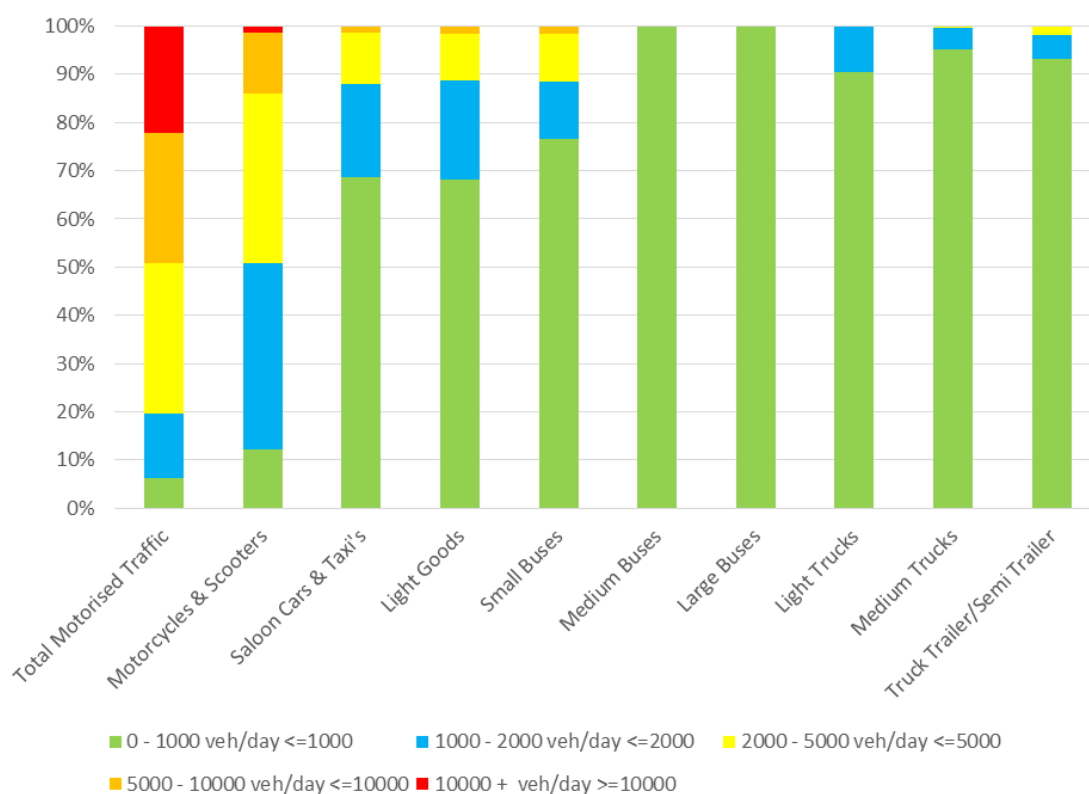


Figure 4-4: Traffic distribution on paved roads per Vehicle Type, 2016

Figure 4-4 confirms that 50% of the travelling concern volumes more than 5,000 vehicles per day (refer Total Motorised Traffic). The figure also shows a breakdown of the vehicle types in the traffic stream, confirming the extensive amount of high volume travel via motorcycles and scooters (Boda-Bodas), followed by saloon cars and taxis, light goods vehicles and small busses. In general, travel exceeding 1,000 vehicles per day is limited to 30% of these three vehicle types, while truck travel exceeding 1,000 vehicles per day is limited to 10% of these vehicle types.

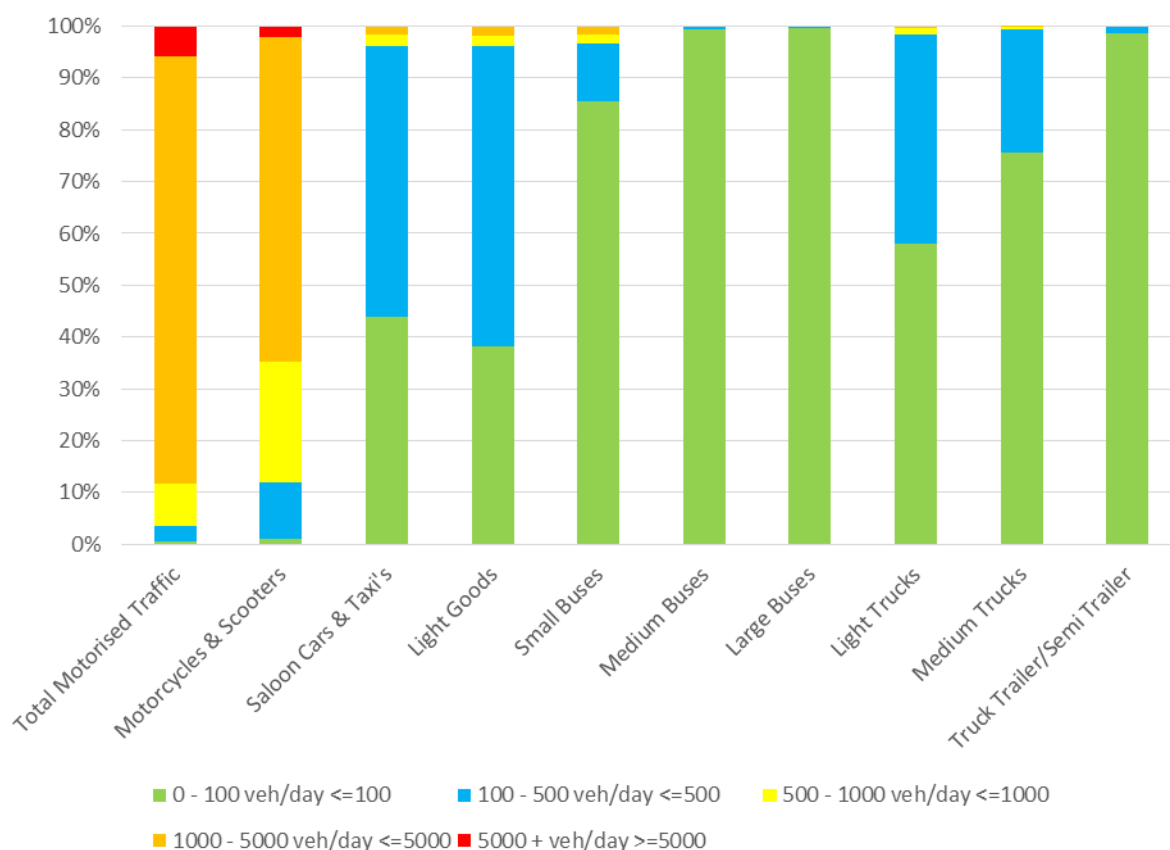


Figure 4-5: Traffic distribution on unpaved roads per Vehicle Type, 2016

For unpaved roads, Figure 4-5 confirms that 85% of unpaved road travel concerns volumes between 1,000 and 5,000 vehicles per day. Similar to paved roads, most of the travel is by motorcycles and scooters. 40% to 60% of travel via saloon cars and taxis, light goods and light trucks are between 100 and 500 vehicles per day. 15 to 25% of small busses and medium truck volumes are between 100 and 500 vehicles per day.

5 Paved Road Network Condition

Summary:

- *The average paved road VCI based on the 2015 visual assessment data is 78%.*
- *Moderate to severe ‘All Cracking’ is noted on 50% of the paved roads.*
- *10% of roads have severe to warning ravelling (surface disintegration) and immediate maintenance is required on these roads.*
- *There is severe to warning roughness on 10% of the road network.*
- *30% of the road network exhibits rutting in moderate to severe state.*
- *Width loss (due to edge break) of moderate to isolated state only occurs on approximately 5% of the network where no shoulders exist.*
- *As for width loss, edge drops are severe to moderate where paved shoulders exist, on 30% of the network.*
- *Overall, the paved road network is in good condition with less than 10% in poor and very poor condition.*

5.1 Background to the VCI

The assessment data, expressing the carriageway condition through the assessment ratings of pavement defects are utilised to calculate a single Visual Condition Index (VCI). This VCI is calculated for each paved road segment (typically of 1 km length) that was surveyed. VCI is a percentage index ranging between 0 and 100; 0 represents a road segment in very poor condition and 100 represents a road segment in very good condition. During this analysis the VCI is used as follows:

- An average network VCI is calculated representing the average condition of the paved road network or any sub-network (eg regional network). This average network VCI is weighted by length and is used to describe / investigate the average condition of the road network / sub-network.

- The VCI is furthermore used to categorise the condition of a paved road segment, the paved road network or sub-network into one of five condition categories. The condition categories adopted are:
 - Very Good = 86% to 100%
 - Good = 71% to 85%
 - Fair = 51% to 70%
 - Poor = 31% to 50%
 - Very Poor = 0% to 30%

5.2 Visual Assessment

The condition of flexible pavements was assessed by visual inspection and by profile measurement equipment. UNRA staff executed paved visual assessments in October / November 2015, and paved roughness measurements in 2015 / first quarter 2016. The Pavement Management System (PMS) of the UNRA RMS was updated with this data following basic data validation.

Figure 5-1 is a schematic representation of the distress items showing the total percentage of paved roads rated as

1  (very good or non-existent), 2  (slight, good or isolated occurrence), 3  (discernible, fair or moderate occurrence), 4  (warning, poor or general occurrence) and 5  (severe, very poor or extensive occurrence). These ratings can indirectly be related to the proportion of the paved road network with  none,  isolated,  moderate,  warning and  severe problems.

- No rut depth measurements were available for 2015 therefore no conclusions can be made regarding this distress. Rut depth was however visually assessed.
- According to international standards (also advocated by World Bank), a “healthy” road network has less than 10% poor and very poor roads. Attention should thus be given to the distresses exceeding this benchmark for warning and severe problems.

According to the visual assessments and automated measurements the following conclusions can be made with regards to the paved road network:

- Cracking
- Moderate to severe ‘All Cracking’ is noted on 50% of the paved roads. The ratings for ‘Wide Cracking’ indicate the development of structural cracking on more than 50% of roads with all cracks. Wide cracking is a particular concern because these cracks indicate the disintegration of the structural layers in the wheel paths. Wide cracks also initiate the formation of rutting. There are moderate to severe ratings for wide cracks on 20% of the paved road network. Preventive measures should be considered as soon as possible to delay further deterioration with the development of secondary distresses.
- Ravelling
- Ravelling is the loss of aggregate material from the road surfacing. In seals, this manifests through stone loss, in asphalts through disintegration of the surfacing.
 - 10% of roads have severe to warning ravelling / surface disintegration and immediate maintenance is required on these roads. Future maintenance of roads with moderate ravelling is crucial since it accounts for 30% of roads which could easily deteriorate to warning condition.
- Potholes
- No roads are in a severe, warning or moderate condition due to the occurrence of potholes. Isolated potholes occur on 15% of the network.
- Roughness (measured)
- There is severe to warning roughness on 10% of the road network. These roads typically require some form of rehabilitation.
 - A substantial portion of the network (roughly 40%) is in a moderate condition regarding roughness. Although some of these roads might not require immediate attention it is crucial to monitor the condition continuously and apply the right treatment at the right time to delay

further deterioration.

- Rutting
- 30% of the road network exhibits rutting in moderate to severe state. Less than 10% has rutting with warning and severe problems, indicating that less than 10% of the network has potential structural problems which manifests as rutting with more than 10mm depth.
- Width Loss
- Width loss (due to edge break) of moderate to isolated state only occurs on approximately 5% of the network where no shoulders exist. Where paved shoulders exist, width loss of severe to moderate states (more than 150mm) is more common on 15% of the network.
- Edge Drops
- As for width loss, edge drops are severe to moderate where paved shoulders exist, on approximately 30% of the network. This indicates to edge drops more than 100mm height.

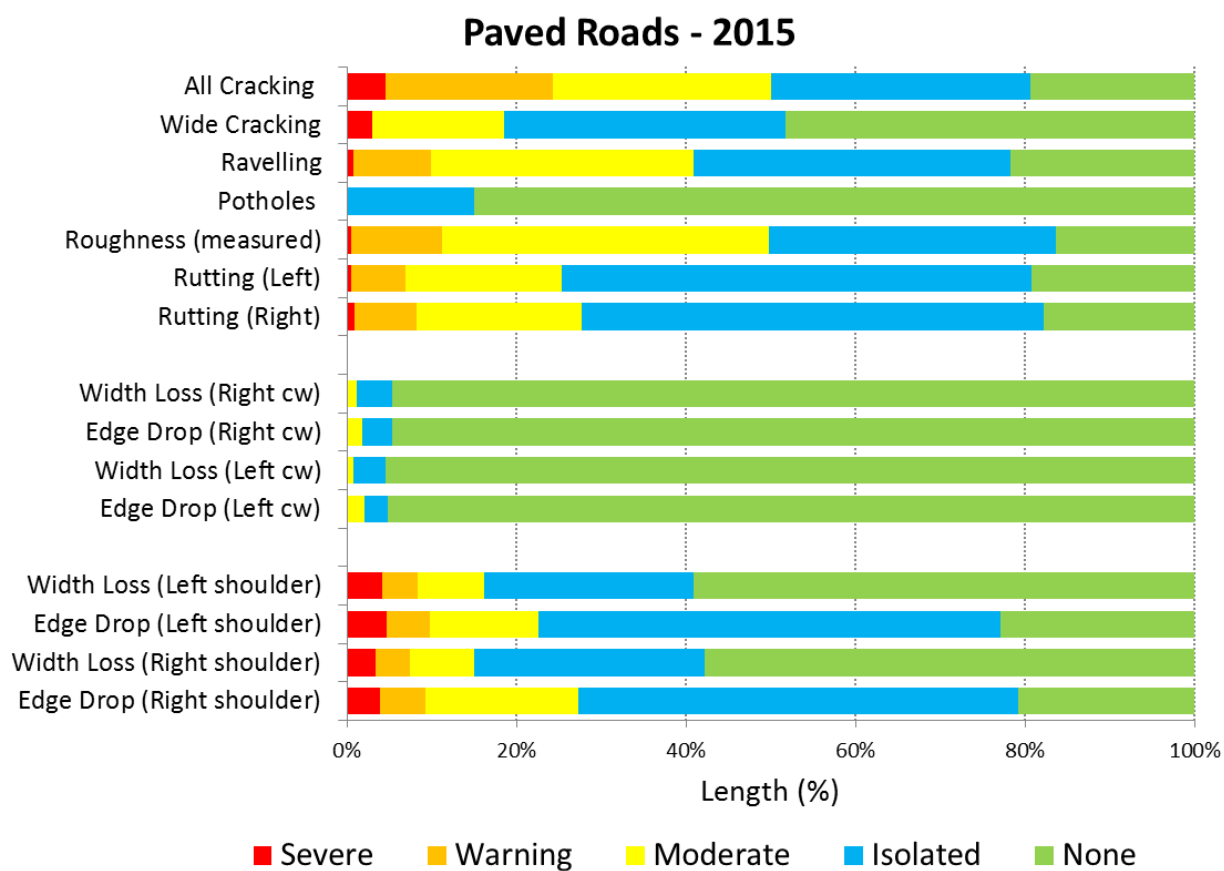


Figure 5-1: Measured values and distress ratings of paved roads, 2015

5.3 Average VCI for Paved roads

The average paved road VCI based on the 2015 visual assessment data is 78%, which places the road network in the “good” condition category. This is a 5 percentage point improvement on the 2009 & 2011 average VCI.

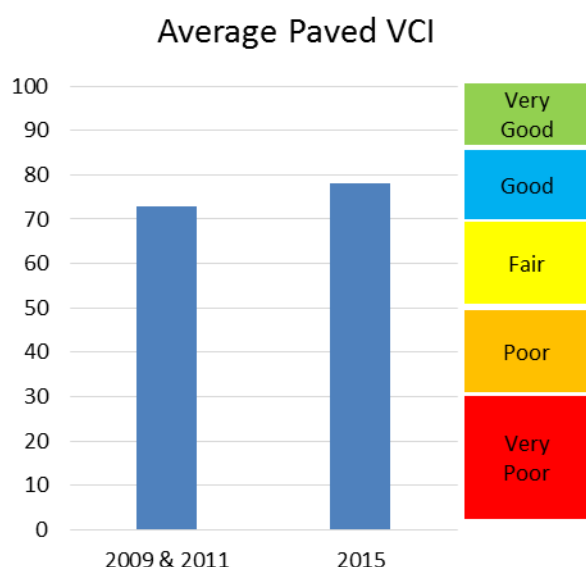


Figure 5-2: Average Paved VCI comparison

5.4 VCI Distribution (by length) for the paved road network

Figure 5-3 and Figure 5-4 show that the percentage of roads in very poor and poor condition has decreased from 12% in 2009 & 2011 to 2% in 2015. The roads in very good condition has however also decreased (with 7%) since the assessment in 2009&2011. The percentage of roads in fair condition is very similar (at 23%, versus 21% previously), while the percentage of roads in the ‘good’ category has increased by 15%.

Overall, the paved road network is in good condition with less than 10% in poor and very poor condition, indicating a low backlog for rehabilitation. However, the 23% of roads in fair condition indicates some backlog for preventive maintenance.

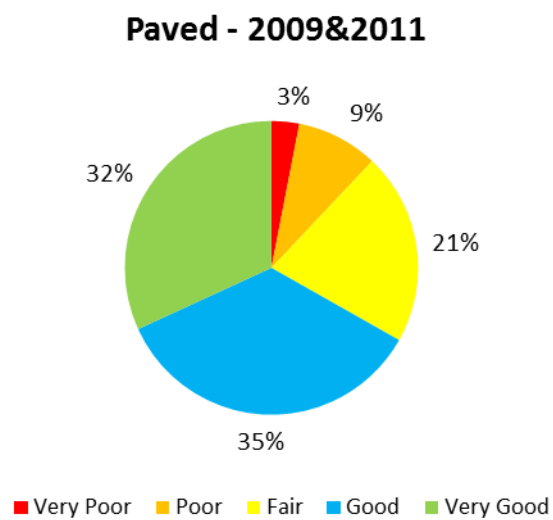


Figure 5-3: Paved VCI distribution for 2009&2011

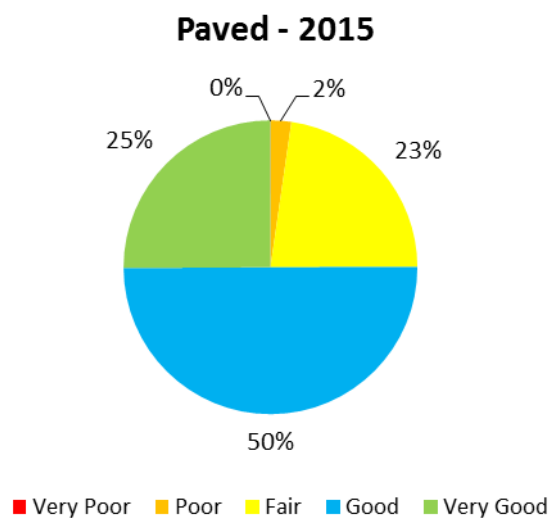


Figure 5-4: Paved VCI distribution for 2015

6 Unpaved Road Network Condition

Summary:

- *The average unpaved road VCI based on the 2015 data is 61% which categorises the road network in the “fair” condition category.*
- *The gravel thickness ratings indicate the need for urgent re-gravelling on 25%.*
- *The roughness ratings on the unpaved network indicates to severe to moderate condition states on 50% of the road network.*
- *Roads with rutting in a moderate to severe condition accounts for almost 50% of the network.*
- *40% of all roads have severe to moderate corrugations.*
- *Erosion is very common with almost 55% of roads showing the occurrence of severe to moderate erosion.*
- *Drainage condition is poor and very poor on 15% of the network.*
- *% of Roads in very poor and poor condition has increased slightly from 24% in 2009 & 2011 to 25% in 2015.*

6.1 Background to VCI

The assessment ratings are utilised to calculate a Visual Condition Index (VCI) similar to the VCI of paved roads. This unpaved VCI is calculated for each unpaved road segment that was surveyed. VCI is a percentage index ranging between 0 and 100; 0 represents a road segment in very poor condition and 100 represents a road segment in very good condition. During this analysis the VCI is used as follows:

- An average unpaved network VCI is calculated representing the average condition of the unpaved road network or any sub-network. This average network VCI is weighted by length and is used to describe / investigate the average condition of the road network / sub-network.

- The VCI is furthermore used to categorise the condition of an unpaved road segment, the unpaved road network or sub-network into one of five condition categories. The condition categories adopted are:
 - Very Good = 86% to 100%
 - Good = 71% to 85%
 - Fair = 51% to 70%
 - Poor = 31% to 50%
 - Very Poor = 0% to 30%

6.2 Visual Assessment

Unpaved visual assessment data was collected in February 2015 and October / November 2015 and the PMS updated with these data.

- | | |
|------------------|---|
| Gravel thickness | <ul style="list-style-type: none"> ■ The gravel thickness ratings indicate the need for re-gravelling. This need is currently very high, with more than 10% having no gravel, another 30% having less than 50mm gravel wearing course and severe exposure of the subgrade. |
| Roughness Rating | <ul style="list-style-type: none"> ■ The roughness ratings on the unpaved network indicates to severe to moderate condition states on 50% of the road network. This will result in high road user costs. |
| Potholes | <ul style="list-style-type: none"> ■ The high occurrence of potholes especially in severe to moderate states (close to 40%) and another 40% in isolated state is indicative of inadequate maintenance. |
| Rutting | <ul style="list-style-type: none"> ■ Roads with rutting in a moderate to severe condition accounts for almost 50% of the network. These roads are in need of blading and re- |

graveling.

- Corrugations ■ 40% of all roads have severe to moderate corrugations. A further 40% has isolated corrugations. Although these roads are in need of blading, there seems to be a need to investigate whether material properties contribute to the formation of corrugations.
- Erosion ■ Erosion is very common with almost 55% of roads showing the severe to moderate occurrence of erosion. There is a need to repair erosion more regularly.
- Drain condition and formation level ■ Drainage condition and formation level indicate the ability of roads to drain water away, transversely underneath and longitudinally next to them. Roads with severe or warning formation levels, would act as a canal or be level with ground level, this is the case for almost 15% of the unpaved road network. Furthermore, drainage condition is poor and very poor on 15% of the network.

The assessment data expresses the condition and characteristics of the wearing course material and the functional condition of the gravel roads as a set of rating scores, rating these as 1  (very good or non-existent), 2  (slight, good or isolated occurrence), 3  (discernible, fair or moderate occurrence), 4  (warning, poor or general occurrence) and 5  (severe, very poor or extensive occurrence). These ratings can indirectly be related to the proportion of the unpaved road network with  none,  isolated,  moderate,  warning and  severe problems.

According to the visual assessments the following conclusions can be made with regards to the unpaved roads of UNRA (refer Figure 6-1):

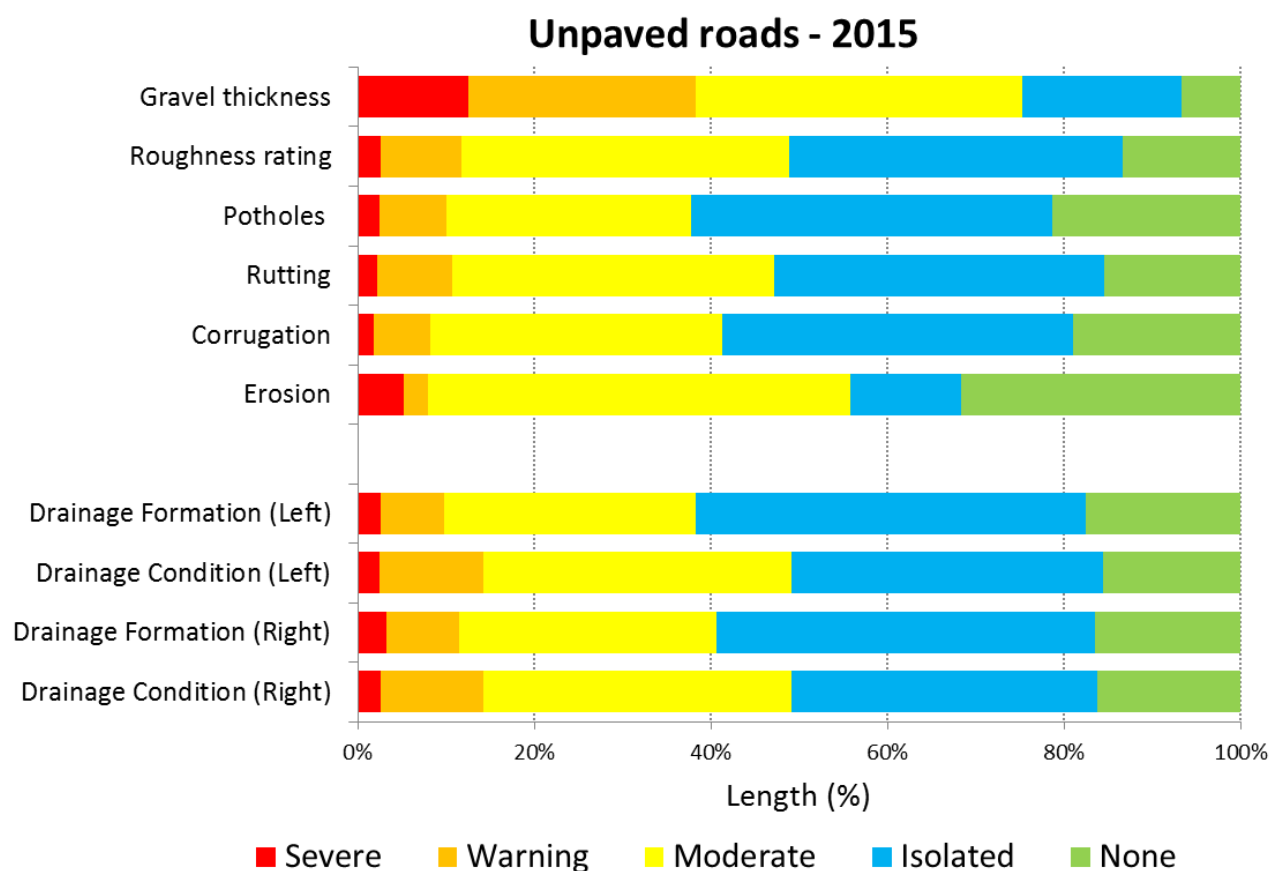


Figure 6-1: Distress, characteristic and condition ratings of unpaved roads, 2015

6.3 Average VCI for Unpaved roads

The average unpaved road VCI based on the 2015 data is 61% which categorises the road network in the “fair” condition category, this is a 1% improvement when compared to the average VCI calculated on the 2009&2011 visual assessment data.

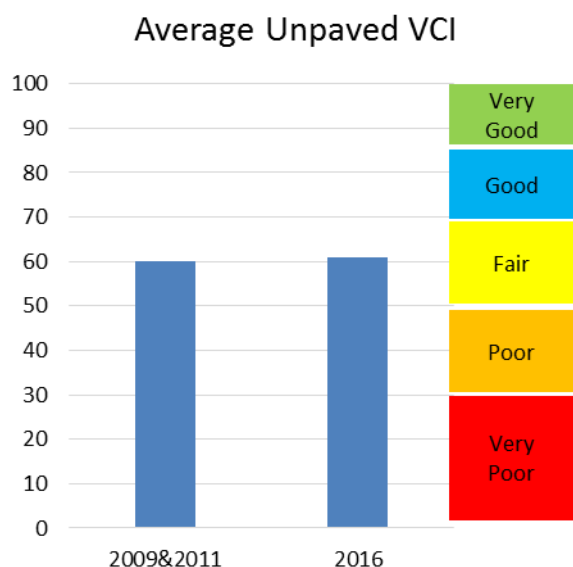


Figure 6-2: Average Unpaved VCI comparison

6.4 Condition distribution graphs for the VCI of Unpaved roads

Figure 6-3 and Figure 6-4 show that the percentage of roads in very poor and poor condition has increased slightly from 24% in 2009&2011 to 25% in 2015. This has thus remained very similar and indicates that a quarter of the network is in urgent need of regravelling. The roads in fair condition has decreased from 50% in 2009&2011 to 43% in 2015, there is therefore an increase in the length of road in good and very good condition (from 26% in 2009&2011 to 32% in 2015). Some regravelling was thus done, or new unpaved roads constructed.

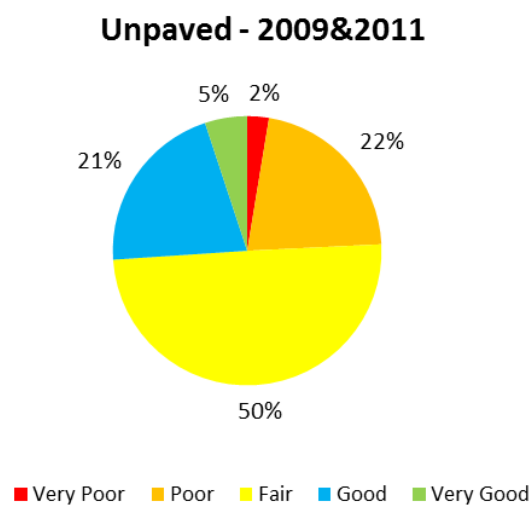


Figure 6-3: Unpaved VCI distribution for 2009&2011

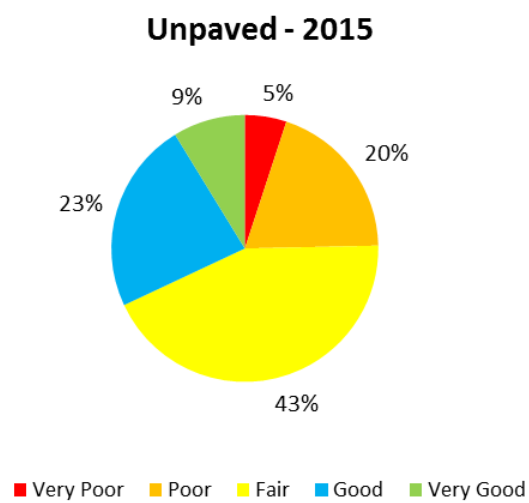


Figure 6-4: Unpaved VCI distribution for 2015

7 The Needs Analysis

Background:

- *HDM-4 pavement performance prediction models are used to model the future performance of the UNRA roads.*
- *The objective of the optimisation function of this LCCA is to preserve the network condition.*

Budget Scenarios:

- *The “Received Budget Scenario” is approximately US\$ 843 million per annum. This allocation is optimised in the analysis to find the optimal investment split between preventive maintenance, rehabilitation and upgrading to paved standards. The maintenance activities investigated are those with identifiable intervention levels¹ and reset values², associated with the pavement performance prediction models.*
- *The “Requested Budget Scenario” of US\$ 1,439 million was also investigated.*
- *The “Technical Needs Scenario” was investigated to determine the immediate need for preventative maintenance, rehabilitation and upgrading to paved standards.*

Output:

- *The optimal allocation of the Received and Requested budgets with regards to the distribution of the funding towards regravelling, reseals, rehabilitation and upgrade to paved standards.*
- *The consequences of the Received Budget on the future road network condition, asset value and vehicle operating costs.*
- *The consequences of the Requested Budget on the future road network condition, asset value and vehicle operating costs.*
- *The current Technical Needs of the road networks; it is the intervention treatments that are required “now”, irrespective of budget. This engineering needs determination provides a base for comparison against planned investment*

¹ The level at which it is necessary to intervene with a treatment, such as reseal, rehabilitation, regravell, etc.

² The value of a pavement performance measure (e.g. distress), after a treatment

levels.

The needs analysis entails the prediction of the future performance of the road network using the current condition and pavement performance prediction models. In the analysis, realistic pavement preventive maintenance, upgrading and rehabilitation repair actions (treatments) are considered for each road analysis segment over a long-term analysis period. The premise is on keeping the roads infrastructure preventively maintained, a world-wide accepted sound infrastructure asset management principle. The effect of each repair alternative is calculated to determine the improvement in road condition and other economic consequences. The preservation of the road network is also taken into account to ensure roads with low traffic are maintained. Furthermore, the result of the analysis is multi-year preventive maintenance and rehabilitation / upgrading plans for different budget scenarios. The main items influencing the needs analysis are:

- Current network condition for paved roads (see Paragraph 5);
- Current network condition for unpaved roads (see Paragraph 6);
- Performance Models – HDM-4 (see Paragraph 7.3);
- Preventive Maintenance and Rehabilitation / Upgrading Treatments and their Costs (see Paragraph 7.4);
- Optimisation Function (see Paragraph 7.5);
- Analysis Scenarios (see Paragraph 7.7).

The road network with condition data described in Chapter 3 was analysed for the purposes of this report and all references to “road network” refers to the road lengths discussed there-in.

7.1 Current Network Condition for paved roads

Visual assessments and roughness measurements were carried out during 2015 to record the extent of defects and performance on paved roads. These assessments are used to determine the current condition of paved roads to serve as the starting point for the future prediction of conditions. See Chapter 5 for a brief discussion on the paved road network condition.

7.2 Current Network Condition for unpaved roads

Visual assessments were carried out during 2015 to record the extent of defects and performance on unpaved roads. Visual assessments determined the current quantity and quality of gravel material on the unpaved road network. The need for maintenance and reg gravel is based on the predicted riding quality that can be achieved due to the characteristics and quality of the wearing course material. See Chapter 6 for a brief discussion on the condition of unpaved roads.

7.3 Pavement Performance Models

7.3.1 Paved roads

To study the long-term consequences of preventive maintenance and rehabilitation actions it is important to use reliable pavement performance prediction models. The pavement performance models of HDM-4 were customised in the dTIMS decision support software for network analysis purposes.

The HDM-4 models attempt to model the complex interaction between vehicles, the environment and the pavement structure and surface. Pavement performance is principally a function of the combined effects of traffic and weather. Traffic loads induce stresses and strains within the pavement layers. The magnitude of these responses depends on the load characteristics and the layer thickness and stiffness. Under repeated loads these stresses and strains cause fatigue in bound materials and deformation of all pavement layers. Weathering and solar radiation causes asphaltic materials to age, become brittle, and more susceptible to cracking and disintegration. Once cracking has been initiated, it increases in both intensity and severity, and eventually potholes are formed. Furthermore, open cracks allow surface water to infiltrate the pavement layers hastening the process of disintegration, reducing the shear strength of the bound materials, and thus increasing the rate of deformation under traffic. The cumulative deformation under wheel loads manifests in wheel path rutting. This in effect increases the pavement roughness, which in turn is further increased by weather and seasonal changes. Pavement roughness, the main indicator of pavement service, is therefore the result of a chain of distress mechanisms and the combination of various modes of

distresses. This complex interaction between various distress types and the environment is reflected in the pavement performance models used in HDM-4.

Five HDM-4 pavement performance models have been customised into the dTIMS software. See Table 7-1 and Figure 7-1 for these distresses and their interaction.

TABLE 7-1: DISTRESSES MODELLED BY HDM-4	
DISTRESS TYPE	DESCRIPTION
All Cracking	The area of cracking greater than 1 mm in width, as a percentage of the total carriageway area.
Wide Cracking	The area of cracking wider than 3 mm in width, as a percentage of the total carriageway area.
Ravelling	The total area ravelled as a percentage of the total carriageway.
Potholing	The total area of open potholes (minimum depth of 25 mm and diameter of 150 mm) as a percentage of the total carriageway.
Rutting	The mean and standard deviation of rut depth.
Road Roughness	Roughness in IRI (International Roughness Index)

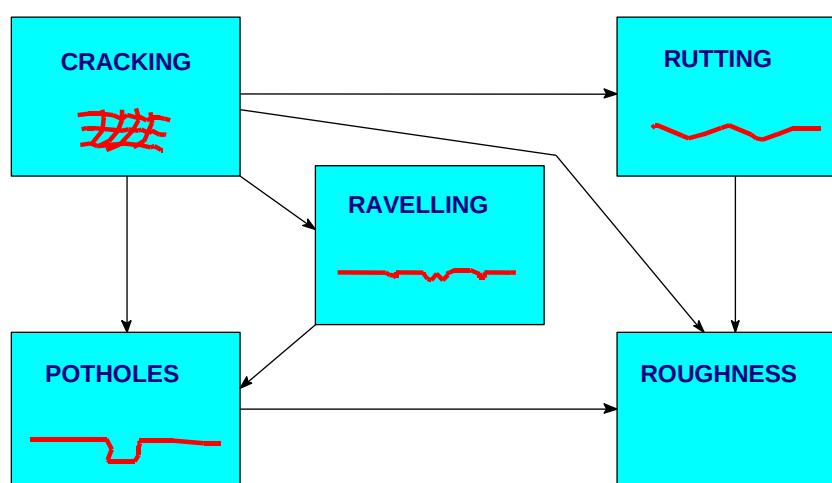


Figure 7-1: Interaction of the distress types as modelled in HDM-4

7.3.2 Unpaved roads

The needs analysis entails the prediction of the future performance of gravel roads using the current condition as the starting point in the gravel road performance models. In the analysis,

realistic maintenance, regravelling and upgrading actions are considered for each road analysis segment over an analysis period. The effect of each repair alternative is calculated as well as the vehicle operating cost associated with the predicted gravel road performance. The result of the analysis is a multi-year list for routine blading maintenance, regravelling (rehabilitation of the wearing course) and upgrades to paved standards for different budget scenarios, whilst minimising the savings in costs for the road users and the road authority. The main items influencing the needs analysis are:

- present condition;
- the HDM-4 pavement performance models;
- maintenance, regravelling and upgrading treatments and their costs;
- vehicle operating cost; and
- the optimisation function.

7.4 Preventive Maintenance and Rehabilitation Treatments

In the analysis, sets of preventive maintenance and rehabilitation strategies (sequence of treatments) were considered for each road segment according to its current condition, predicted future conditions and benefits. Table 7-2 describes the treatments, their triggers and costs. These unit costs are financial unit costs and were provided by UNRA as derived from recent contracts.

TABLE 7-2: TREATMENTS AND TRIGGERS		
TREATMENT	COST 2015 IN US\$	TRIGGER
Reseal	10 / m ²	All cracks > 2% and wide cracks ≤ 15% and roughness ≤ 4.2 IRI and rutting ≤ 20mm and AADT ≥ 4000 or Wide cracks between 5% and 10% and AADT < 4000 and roughness ≤ 4.2 IRI and rutting ≤ 20mm
Reseal Light	8 / m ²	All cracks > 1% and wide cracks ≤ 5% and roughness ≤

		4.25 IRI and rutting $\leq 20\text{mm}$ and AADT < 4000
Light Rehabilitation	35 / m ²	Wide cracks $> 10\%$ and condition $> 45\%$ and roughness ≤ 6.6 IRI and rutting $< 20\text{mm}$ or Wide cracks $\leq 10\%$ and roughness > 5.5 but < 6.6 IRI and rutting $< 20\text{mm}$ or Roughness ≤ 5.5 and wide cracks $> 10\%$ and rutting $< 20\text{mm}$
Heavy Rehabilitation	110 / m ²	Roughness > 5.5 IRI and wide cracks $> 10\%$ or Roughness > 6.6 IRI or Rutting $> 20\text{mm}$ or Overall pavement condition very poor
Regravel	13,500 / km	The option to regravel the road is triggered when the wearing course thickness is less than 30mm.
Upgrade to standard paved road	740,000 / km	Traffic > 200 AADT and economic merit, i.e. savings in road user costs exceed the cost of the upgrade. Total transport costs (TTC) must be positive (refer 7.5.2).

7.5 Optimisation Function

In this analysis the maintenance plan was derived with the overall objective to maximise the condition of the road network through the Area-Under-the-condition-Curve (AUC) approach. The best possible condition under the budget constraint is thus aimed for. For unpaved roads, life cycle Total Transportation Costs (TTC) were calculated to identify feasible links for upgrading to paved standards. This ensures feasible recommendations with regards to proposed upgrades to paved standards.

7.5.1 Area-under-the-condition-curve (AUC)

The area-under-the-curve benefit is calculated by summing the present value of the difference between the condition index resulting from the repair strategy (combination of repair actions

over the analysis period) and the condition index for the Do-Nothing alternative for each year in the Analysis Period. The condition index is a combination of the pavement performance models of HDM-4, namely cracking, potholes, roughness, rut depth and gravel thickness. The area-under-the-condition curve benefit calculations are weighted by traffic (AADT). Figure 7-2 illustrates the method of calculating this benefit.

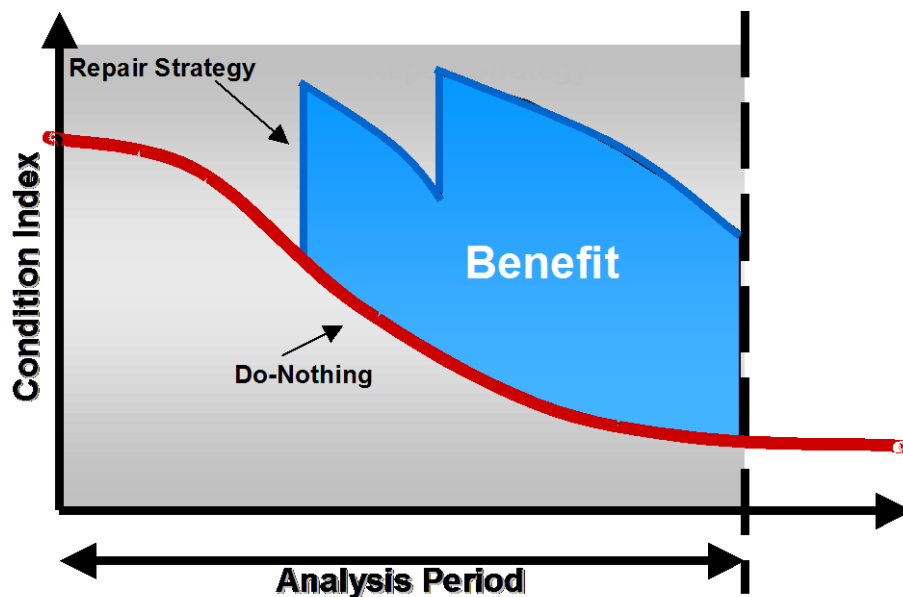


Figure 7-2: Area-Under-the-Condition Curve

The following equation represents the benefit calculation in year "i" for the area-under-the-condition curve approach.

$$\text{Benefit (i)} = \text{AADT (i)} * (\text{Str_Cond(i)} - \text{DN_Cond(i)})$$

Where: i = the year in the Analysis Period

Benefit (i) = Benefit of the Strategy for year i

AADT (i) = AADT on the Element in year i

Str_Cond(i) = Value for the repair strategy's condition index in year i

DN_Cond(i) = Value for the do-nothing alternative's condition index in year i

For each strategy a Condition Index (CIND) is calculated representing the condition of the road segment over time. The area underneath the curve is the "distance" between the condition index of the Do Nothing strategy and the Repair strategy, totalled for all analysis

years and weighted by traffic. Any strategy that improves the condition of the pavement will thus have a positive area underneath the curve because it is an improvement in comparison to the Do Nothing Strategy. Roads carrying high traffic receive a higher benefit than those with lower traffic. This method is used on networks where there are also roads with traffic volumes that are low.

7.5.2 Total transport cost (TTC)

Another method of optimisation is the life-cycle total transportation costs (TTC) that includes both the costs of maintaining a road network and the costs of operating vehicles upon it. Increased expenditure by the roads authority will lead to reduced cost for road users but there is theoretically a level of expenditure by the road authority at which the total transport costs are optimised. This concept is graphically illustrated in Figure 7-3. As the maintenance level (and cost) increases, the vehicle operating costs decrease. However, the increased spending on maintenance is not necessarily proportional to the savings in user costs resulting in an optimal maintenance scenario where the total transport cost to society can be minimised. Ideally a road authority should maintain a road system at the maintenance level associated with the minimum TTC.

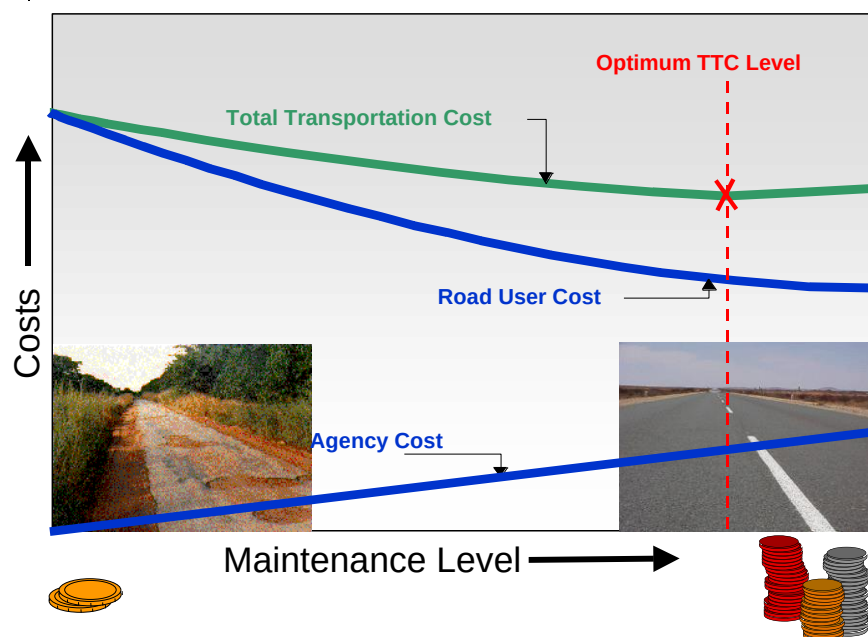


Figure 7-3: Illustration of Optimum Total Transport Costs

If this method is used for optimisation, maintenance strategies are selected to ensure the combined sum of the costs to the roads authority and road users are minimised. The savings in road user costs should always exceed the cost of the maintenance or repair strategy otherwise no economic merit exists to perform such a maintenance strategy. This minimisation of TTC therefore produces the lowest economic cost. *However, low traffic roads do not always have sufficient economic justification for maintenance due to limited savings in road user costs. This optimisation method is not recommended for UNRA because roads without economic merit for maintenance will deteriorate beyond repair, and accessibility will be compromised.* It is for this reason that the road network of UNRA is optimised using the AUC method, but the TTC functions were added to determine which unpaved roads have economic merit for upgrading to paved standards.

7.6 Committed Projects

Appendix A contains the committed projects that were included in the analysis. Some projects were committed without a cost. These projects have been completed in 2016, the first analysis year.

7.7 Budget scenarios

The results of the analysis is highly dependent on the funding available for the maintenance of the network and all results are thus expressed as the consequence of a funding / budget scenario.

The funding scenarios discussed and analysed in this report concern the budget values for the following treatments only:

- Preventive maintenance (light reseal / reseal) of paved roads.
- Rehabilitation of paved roads (light / heavy).
- Regravelling of unpaved roads.
- Upgrading of unpaved roads to paved standards.

Other budget items, such as routine maintenance, capacity and geometric improvements, bridge maintenance and repairs, salaries, overheads, etc., are not analysed in this report and these requirements should be added to any budget needs that calculated during this analysis.

The total budget of UNRA is US\$ 843 million for 2016. Another, but higher constrained budget scenario and a Do Nothing scenario were analysed to investigate the consequences of various funding levels. A Technical Needs scenario is included to determine the immediate need (backlog). All budgets investigated are discussed in Table 7-3.

TABLE 7-3: FUNDING SCENARIOS INVESTIGATED			
FUNDING SCENARIO		DESCRIPTION	FUNDING LEVEL
Constrained funding scenarios	“Received Budget” US\$ 843 million	This scenario investigates the future network performance if an annual investment of US\$ 843 million is made for reseals, rehabilitation, regravelling and upgrading to paved standards.	US\$ 843 million per annum
	“Requested Budget” US\$ 1,439 million	This scenario investigates the future network performance if an annual investment of US\$ 1,439 million is made for reseals, rehabilitation, regravelling and upgrading to paved standards.	US\$ 1,439 million per annum
	“Do Nothing” Budget	This scenario was analysed to determine the condition and deterioration of the network if no funding was available. No committed projects were analysed as part of this scenario.	US\$ 0 million per annum

“Technical Needs Budget”	This scenario selects the first and earliest treatment that is triggered for each road segment, according to the intervention measures adopted for UNRA. This scenario is used to determine the funding level that is currently required to remove the backlog with regards to preventive maintenance, rehabilitation and upgrading to paved standards.	Funding need for the year 2016 is determined by the immediate technical needs of the road network.
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7.8 Other analysis inputs

UNRA provided a traffic baseline growth rate of 6% to be used for traffic growth on all roads in the network. This was considered the best estimate currently, given a lack of country wide growth rate data.

The analysis and planning period of the analysis was taken as 10 years.

A discount rate of 8% was used in the analysis.

8 Immediate Technical Needs (Backlog)

Summary:

This scenario determined the funding level for UNRA roads to perform all the preventive maintenance, rehabilitation and upgrading to paved standard measures, needed from an engineering point of view, within the first analysis year.

The immediate technical needs for roads in the UNRA network are:

- *US\$ 299 million is needed for light and heavy rehabilitation of 946 km paved roads.*
- *US\$ 76 million is needed to reseal 947 km of paved roads to deter further deterioration (this includes light reseal).*
- *US\$ 5,567 million is required for upgrading of 8,687 km of unpaved roads to paved standards.*
- *US\$ 2 million is needed for regravelling 510 km of unpaved roads.*

The Technical Needs scenario determines the required budget in 2016 in order to reinstate the road network to a good condition. All backlog is removed immediately. The analysis determines which roads require immediate intervention of light reseal, reseal, rehabilitation, upgrade to paved standards or regravelling, and the analysis subsequently calculates the associated costs. The costs are then expressed as the Technical Needs Budget and it gives the road authority an estimate of the immediate backlog with regards to the maintenance / rehabilitation / upgrading activities investigated in this report.

This scenario will have the following consequences:

- The backlog of paved roads in a poor to very poor condition will be removed,
- all unpaved roads with justifiable economic benefits will be upgraded to paved standards, and
- all unpaved roads with insufficient gravel material will be regravelled.

The optimisation analysis selected the most cost-effective alternatives for 2016 according to the most recent condition surveys that were completed on the network:

- The immediate need for paved roads is US\$ 375 million
 - US\$ 299 million is needed for light and heavy rehabilitation: This is 946 km of road where the roughness and/or structural cracking have developed beyond preventive maintenance; structural repairs are needed to improve the functional levels.
 - US\$ 76 million is needed for reseals: Approximately 947 km of paved roads should be resealed as a preventative measure to deter further deterioration.
- The immediate need for unpaved roads is US\$ 5,569 million ($\pm$ US\$ 5,5 billion)
 - US\$ 5,567 million is required for upgrading of 8,687 km of unpaved roads to paved standards.
 - US\$ 2 million is needed for regravelling: In total 510 km of unpaved roads require replacement of the gravel wearing material. The road user costs on gravel roads with no wearing course material are normally excessive.

Figure 8-1 shows the funding level required to remove the backlog and Figure 8-2 shows the length of road affected by this funding scenario.

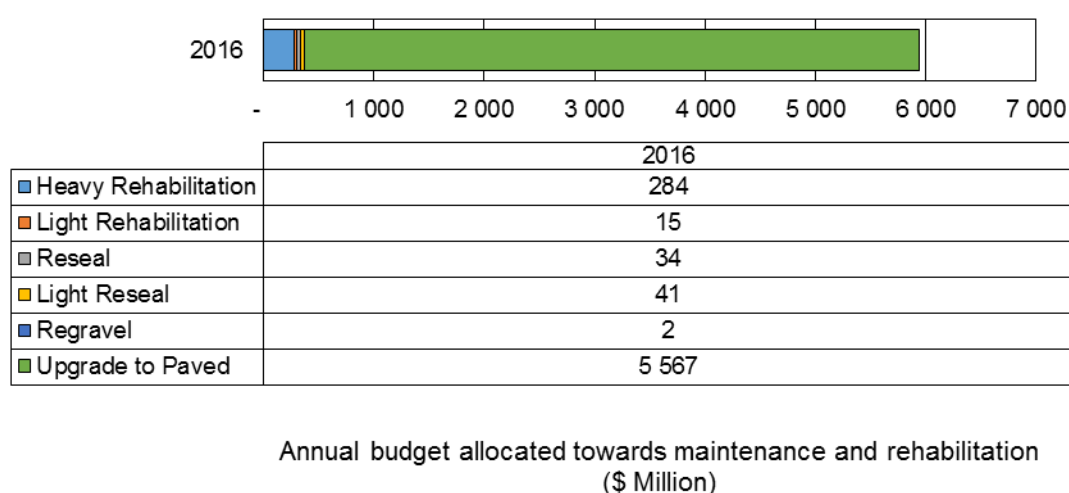


Figure 8-1: Immediate Technical Needs 2016 (cost, US\$ million)

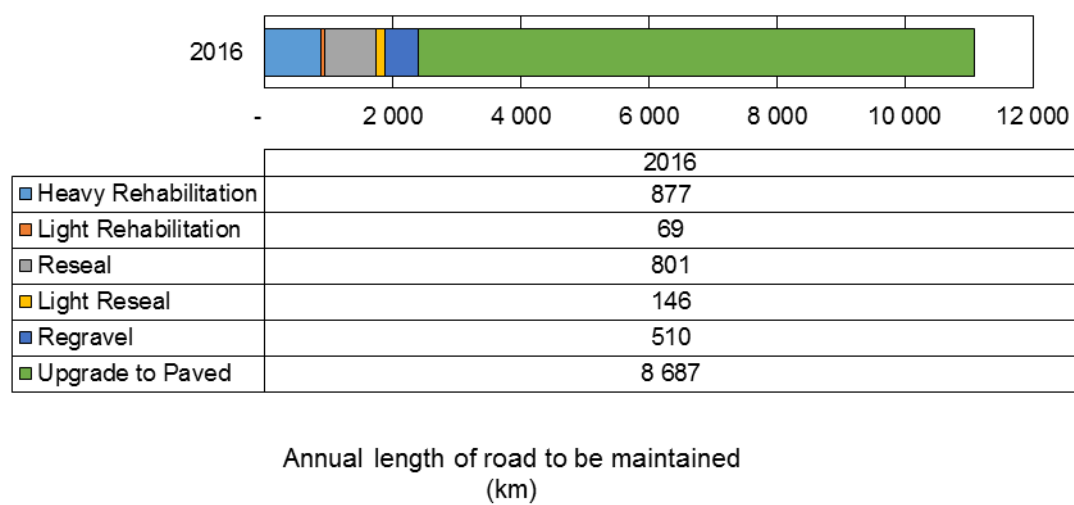


Figure 8-2: Immediate Technical Needs 2016 (length, km)

9 Received Funding Scenario

Summary:

This scenario determined the optimised funding allocation of the received budget towards preventative maintenance, rehabilitation and upgrading to paved standards. This is a constrained funding scenario with US\$ 843 million available of which US\$ 753 million is allocated to committed projects in the first year of the analysis.

The following results are relevant

- *US\$ 138 million is available annually for light and heavy rehabilitation of paved roads. Over a five year period it is possible to rehabilitate 1,536 km.*
- *US\$ 16 million is available annually for reseals. 1,411 km can be resealed in five years.*
- *US\$ 649 million is available annually for upgrading of unpaved roads to paved standards. Over a five year period it is possible to upgrade 5,042 km.*
- *US\$ 40 million is available annually for regravelling. 13,825 km can be regravelled over a five year period.*

The Received Funding scenario determines the optimum allocation of the funds available. This is a constrained funding scenario with US\$ 843 million per annum of which US\$ 753 million is allocated to committed projects in the first year of analysis. Therefore the optimisation of only US\$ 90 million is considered for the first year. Table 9-1 shows the fund allocations towards committed projects for paved and unpaved roads as well as the funds available for optimisation from 2016 up to 2019.

TABLE 9-1: COMMITTED FUNDS AND FUND ALLOCATIONS AVAILABLE FOR OPTIMISATION

YEAR	COMMITTED PAVED (US\$ MILLION)	COMMITTED UNPAVED (US\$ MILLION)	FUNDING AVAILABLE FOR OPTIMISATION
2016	263	490	90

2017	257	497	88
2018	96	342	405
2019	12	40	791

Figure 9-1 shows the cost allocation towards different maintenance and rehabilitation strategies from 2016 to 2020 and Figure 9-2 shows the length of road affected by the different strategies. It should be noted that the committed projects are included in these figures.

The optimisation analysis selected the most cost-effective alternatives according to the most recent condition surveys that were completed on the network:

- The optimisation analysis determined that on average 18% (US\$ 154 million) should be allocated on an annual basis towards paved roads over a five year period.
 - On average US\$ 138 million of the US\$ 154 million should be invested towards the annual rehabilitation of roads and the remaining US\$ 16 million should be used for resealing.
 - In total over the five year period 1,536 km of road could be rehabilitated and 1,411 km could be resealed.
- On average 82% (US\$ 689 million) of the received budget should be allocated annually to unpaved roads.
 - US\$ 649 million should be used for the yearly upgrading of unpaved roads to paved standard and the remaining US\$ 40 million should go towards regraveling.
 - In total over the five year period 5,042 km of road could be upgraded to paved road standards and 13,825 km could be regravelled.

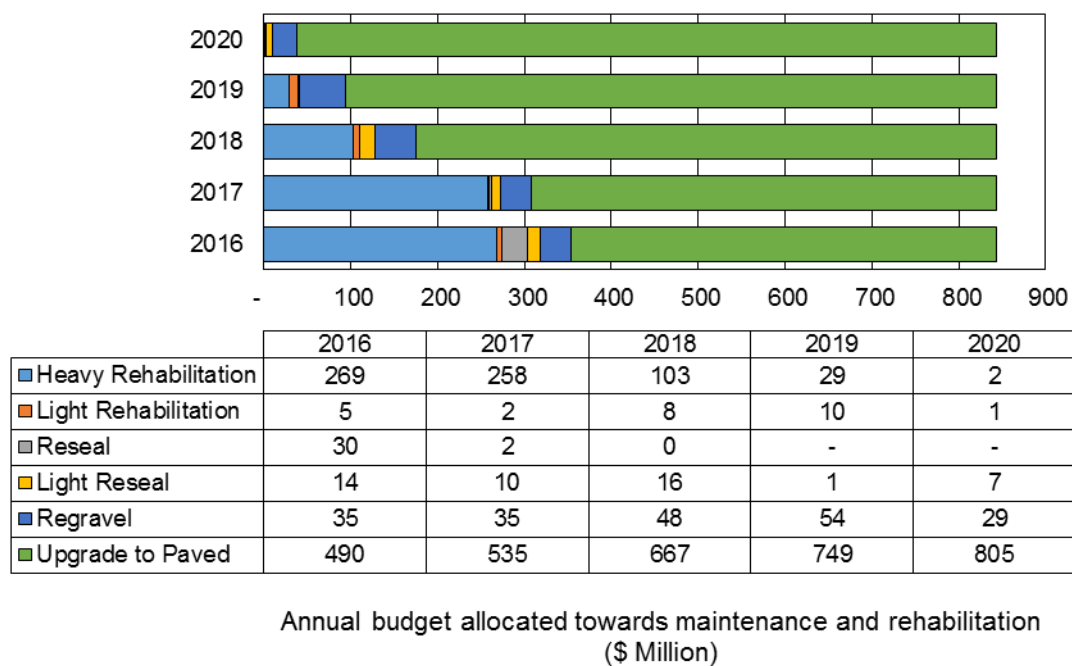


Figure 9-1: Received Funding Scenario optimised allocation (cost, US\$ million)

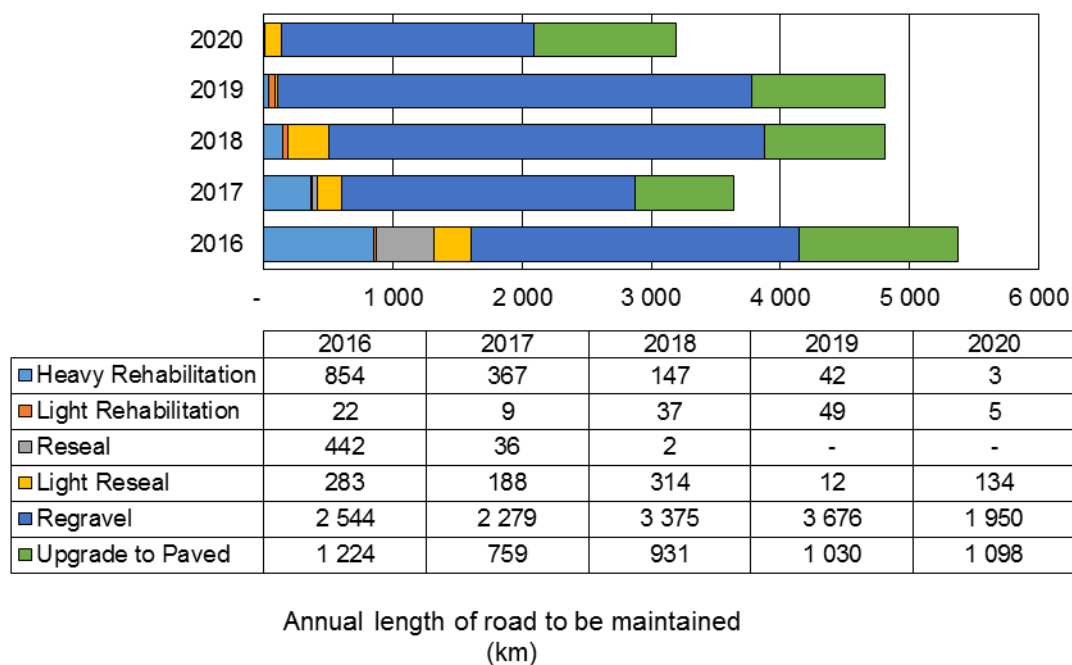


Figure 9-2: Received Funding Scenario optimised allocation (length, km)

10 Requested Funding Scenario

Summary:

This scenario determined the optimised funding allocation of the requested budget towards preventative maintenance, rehabilitation and upgrading. This is a constrained funding scenario with US\$ 1,439 million available per annum of which US\$ 753 million is allocated to committed projects in the first year of the analysis.

The following results are relevant

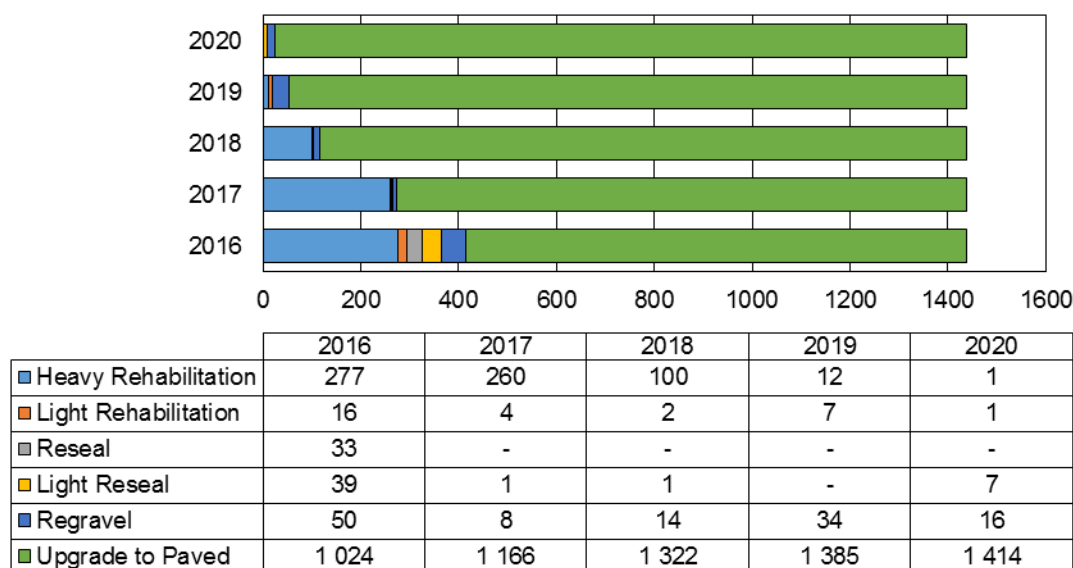
- *US\$ 136 million is available annually for light and heavy rehabilitation of paved roads. Over a five year period it is possible to rehabilitate 1,536 km.*
- *US\$ 17 million is available annually for reseals. 1,432 km can be resealed in five years.*
- *US\$ 1,262 million is available annually for upgrading of unpaved roads to paved standards. Over a five year period it is possible to upgrade 9,526 km.*
- *US\$ 24 million is available annually for regravelling. 7,867 km can be regravelled over a five year period.*

The requested funding scenario of US\$ 1,439 million was analysed and the cost optimisation is shown in Figure 10-1. Figure 10-2 shows the length of road that can be maintained or rehabilitated with the requested funding. It should be noted that the committed projects were included in this analysis. Table 10-1 summarises the fund allocations towards committed projects for paved and unpaved roads as well as the funds available for optimisation from 2016 up to 2019.

TABLE 10-1: COMMITTED FUNDS AND FUND ALLOCATIONS AVAILABLE FOR OPTIMISATION

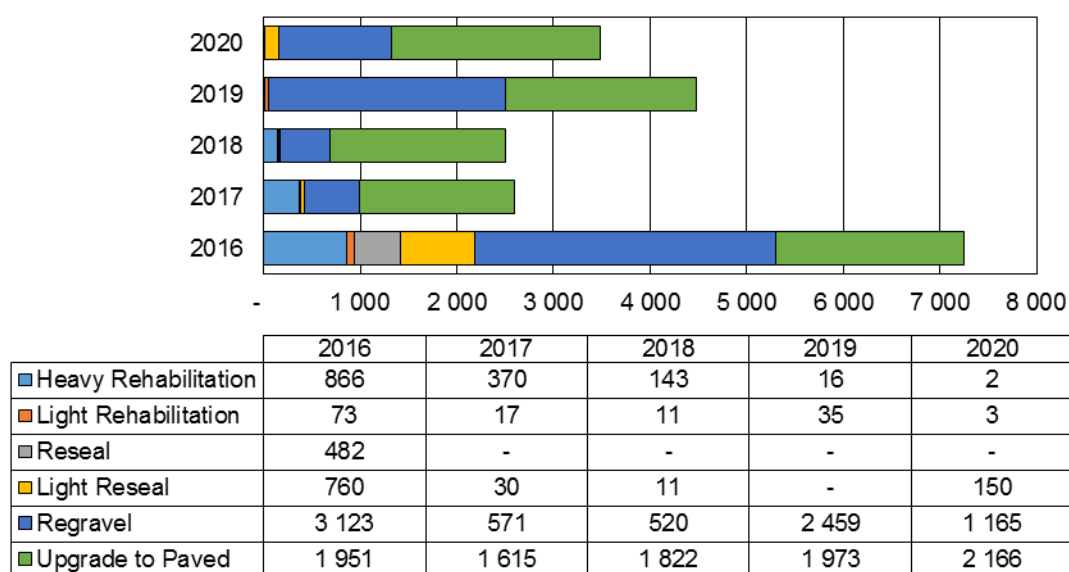
YEAR	COMMITTED PAVED (US\$ MILLION)	COMMITTED UNPAVED (US\$ MILLION)	FUNDING AVAILABLE FOR OPTIMISATION
2016	263	490	686

2017	257	497	684
2018	96	342	1,001
2019	12	40	1,387



Annual budget allocated towards maintenance and rehabilitation
(\$ Million)

Figure 10-1: Requested Funding Scenario optimised allocation (cost, US\$ million)



Annual length of road to be maintained
(km)

Figure 10-2: Requested Funding Scenario optimised allocation (length, km)

The optimisation analysis selected the most cost-effective alternatives according to the most recent condition surveys that were completed on the network:

- The optimisation analysis determined that on average 11% (US\$ 152 million) should be allocated on an annual basis towards paved roads over a five year period.
 - On average US\$ 136 million of the US\$ 152 million should be invested towards the annual rehabilitation of roads and the remaining US\$ 16 million should be used for resealing.
 - In total over the five year period 1,536 km of road could be rehabilitated and 1,432 km could be resealed.
- On average 89% (US\$ 1,286 million) of the Requested Budget should be allocated to unpaved roads annually.
 - US\$ 1,262 million should be used for the yearly upgrading of unpaved roads to paved standards and the remaining US\$ 24 million should be allocated towards regraveling.
 - In total over the five year period 9,526 km of road could be upgraded and 7,837 km could be regravelled.

11 Comparison of Different Funding Levels

Summary:

- *The Received Funding Scenario is sufficient to maintain and improve the condition of the paved road network.*
- *This current funding level is sufficient to keep roads in a “poor” and “very poor” condition below 10%.*
- *The optimum fund allocation towards paved roads is similar for the Received and Requested Funding scenarios.*
- *Budget allocations of higher funding levels would favour the maintenance and upgrading of unpaved roads.*
- *For a funding level below US\$ 800 million the distribution of funds is towards committed projects which mainly account for rehabilitation of paved roads and the upgrading of unpaved roads.*
- *Over a period of 10 years the average gravel thickness will remain around 80 mm for both the Received and Requested Funding Scenarios.*
- *On average the Received Budget will regravell 43% more length of road than the Requested Budget and the Requested Budget will upgrade 47% more road length than the Received Budget.*

11.1 Split between Paved and Unpaved roads

Figure 11-1 shows the allocation towards Paved and Unpaved roads for the three funding levels. It is clear from the graphs that all 3 budget scenarios allocated the same amount to paved roads. Therefore, as the budgets increase the funding allocation towards unpaved roads increase.

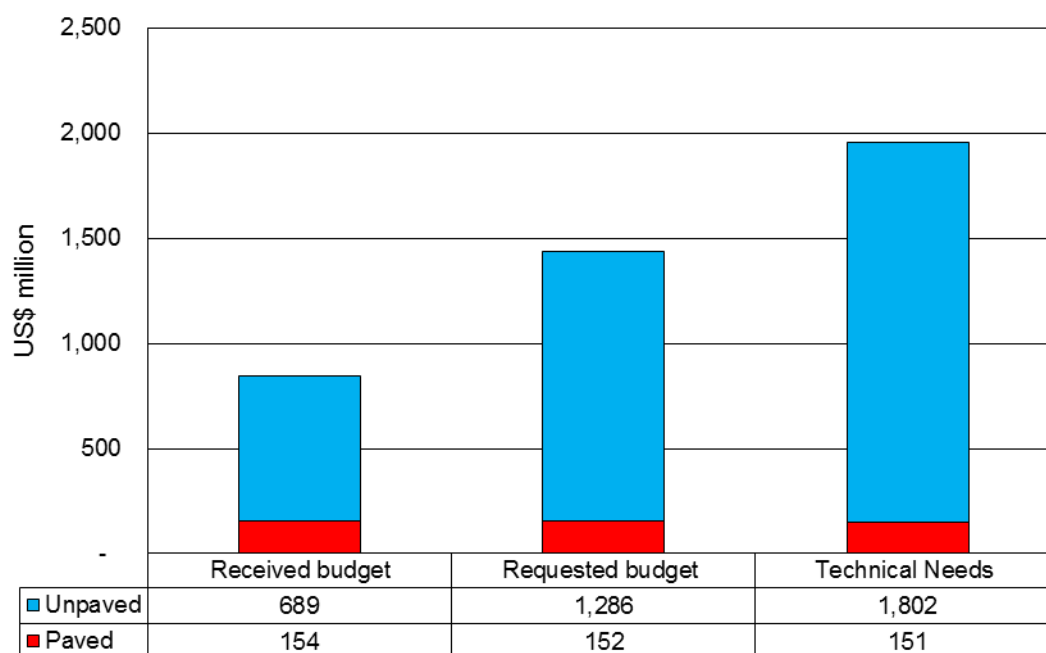


Figure 11-1: Funding split between paved and unpaved roads for three budget scenarios

11.2 Predicted Condition of Paved Roads

During the optimisation analysis, a predicted condition index (CIND) is calculated using the performance models for rut depth, roughness, cracking and potholes. This index has a rating scale of 0% (very poor) to 100% (very good) and is used to provide management with decision support with regards to the future predicted condition under the different funding scenarios. The predicted network condition in Figure 11-2 is the average predicted condition for the paved road network, weighted by the length of the roads.

The predicted condition for the Received, Requested and Technical Needs Scenarios are similar over the next ten years. The current condition of the road network is high and the Received Funding level is sufficient to maintain and even improve the network condition. Therefore any funding scenario such as the Requested and Technical Needs scenarios with higher funding levels will direct additional funds towards the improvement or upgrading of unpaved roads. This can be seen in Figure 11-1 where the funding allocated towards paved roads remains relatively constant for all funding scenarios and the allocation towards unpaved roads increases as more funds become available.

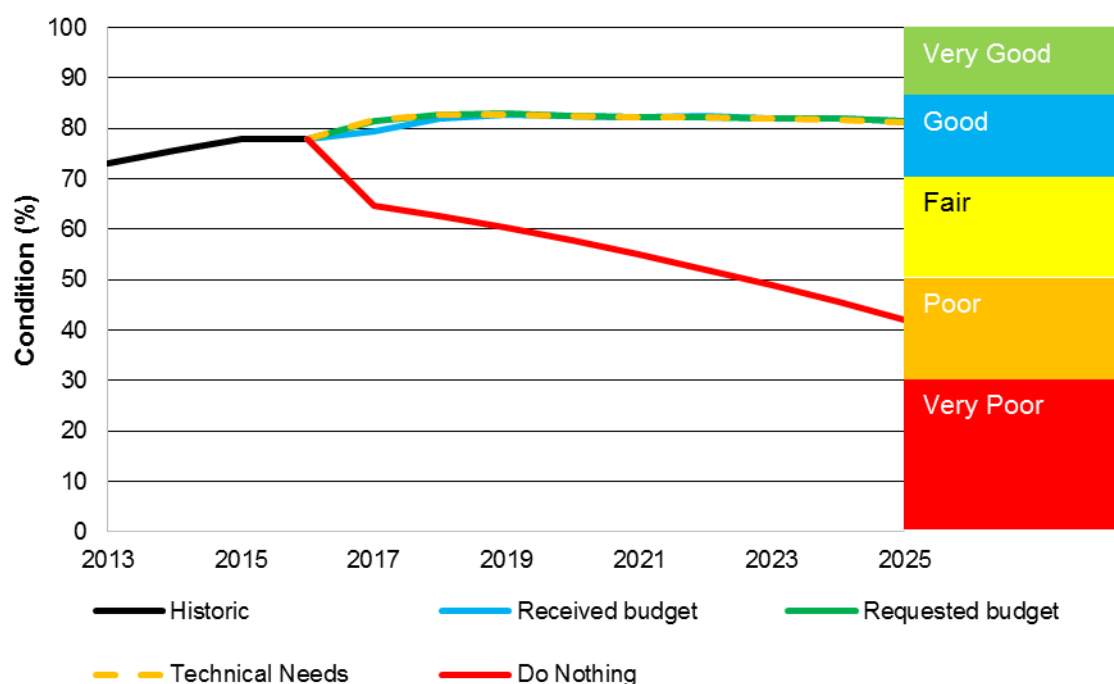
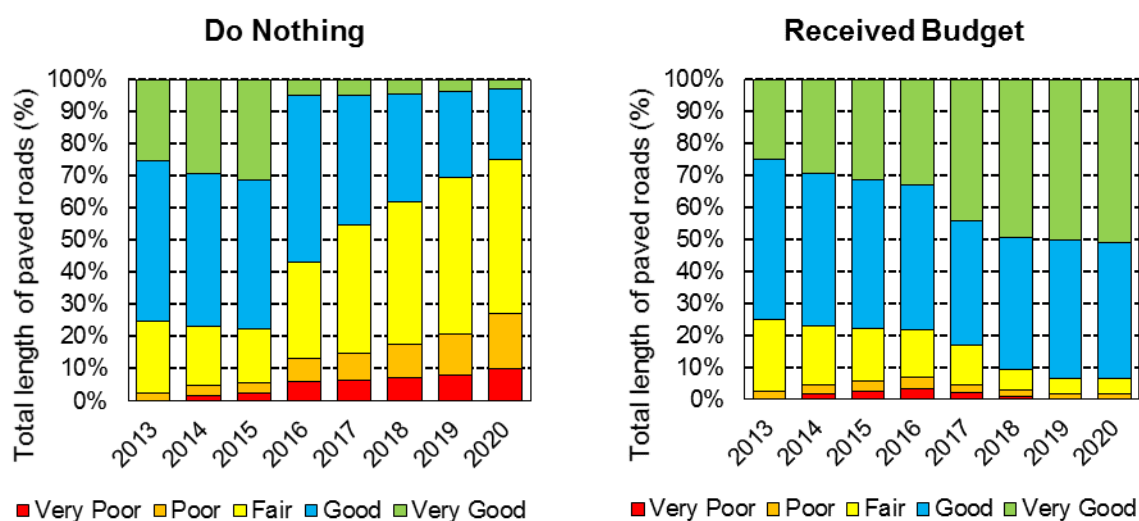


Figure 11-2: Predicted condition for paved roads

Figure 11-3 shows the distribution of the predicted network condition into five condition categories. From an engineering point of view; roads in a poor to very poor condition require expensive repair measures such as rehabilitation to restore the functionality of the roads. Ideally this proportion should be kept at a level not more than 10% of the total road length, which is also advocated by World Bank.



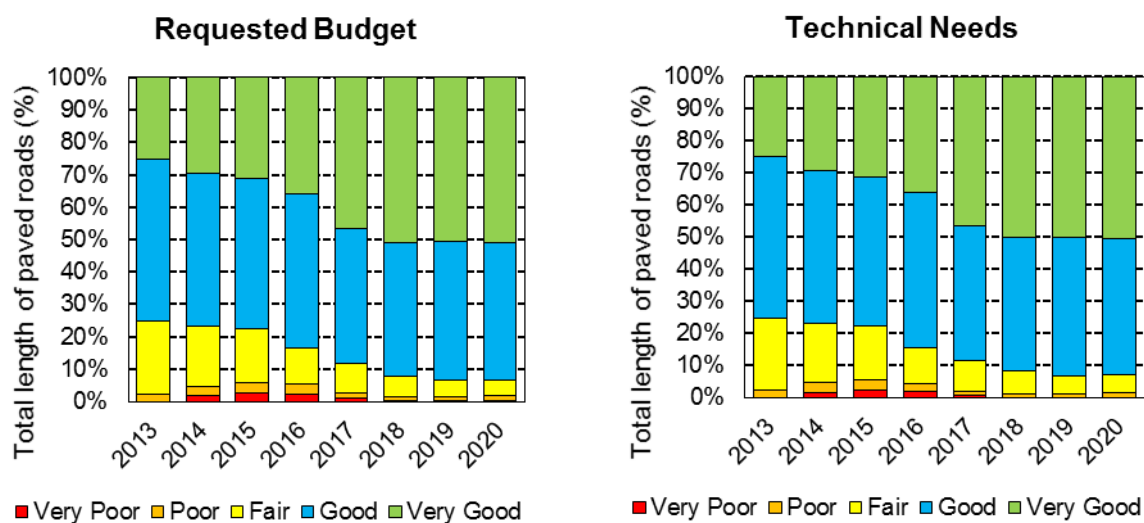


Figure 11-3: Predicted condition distribution for paved roads

Figure 11-4 shows the predicted impact of annualised spending levels on the overall condition of the paved road network. The predicted condition after five years is shown in blue and the predicted condition after ten years is shown in red. The committed projects were also included in this analysis. For a funding level below US\$ 800 million the distribution of funds only goes towards committed projects which mainly account for rehabilitation of paved roads and the upgrading of unpaved roads to paved standards. Because these are expensive treatments the impact that it has on the condition graph is almost insignificant. For funding levels below US\$ 800 million no maintenance on roads is possible and deterioration of roads in a “fair” condition could result in a large backlog in the future.

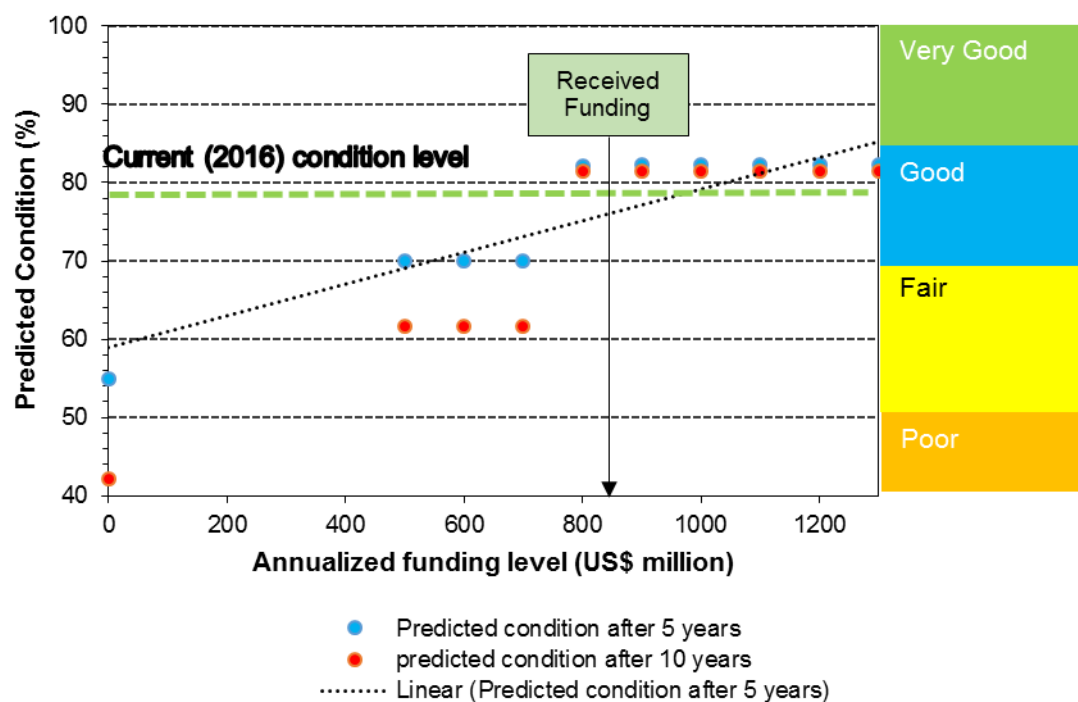


Figure 11-4: Predicted condition versus expenditure for paved roads

Figure 11-5 shows the impact of annualised funding on the predicted proportion of roads in a poor and very poor condition. As previously mentioned funding levels below US\$ 800 million will not be sufficient as all available funds will go towards committed projects. This will increase the portion of the network that is in a “poor” to “very poor” condition. The Received Funding level is sufficient to keep roads in a “poor” and “very poor” condition below 10%.

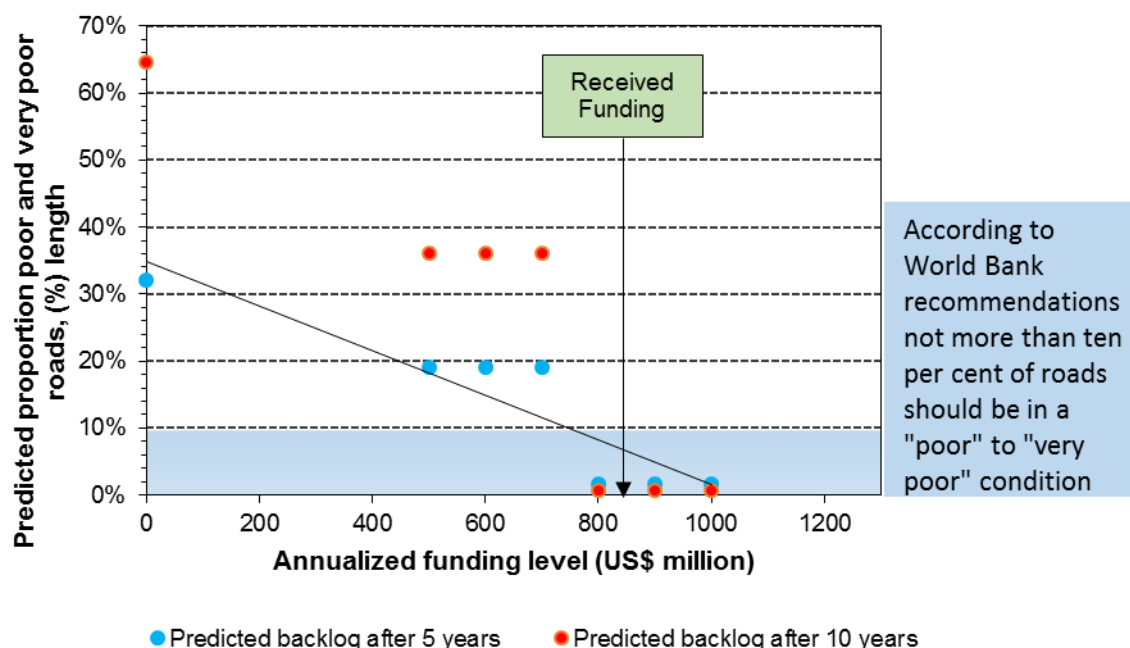


Figure 11-5: Funding levels versus the portion of roads in a “poor” and “very poor” condition

11.3 Predicted Gravel Thickness of Unpaved Roads

Gravel thickness is used to measure the health of the unpaved road network. The roughness model of unpaved roads is dependent on the quality and quantity of the gravel wearing course material which influences the performance of the unpaved roads. During the optimisation process, regravelling is recommended as a preventive measure to ensure unpaved roads are passable and road user costs are kept at acceptable levels. Under constrained funding scenarios the optimisation analysis favours paved roads initially to ensure heavily trafficked roads are maintained. Once the condition of paved roads is at acceptable levels, more funding is directed towards unpaved roads.

All three funding scenarios follow the same trend. Over a period of 10 years the average gravel thickness is around 80 mm. As the funding level increases, the allocation between regravelling and upgrading leans toward the upgrading of unpaved roads to paved standards.

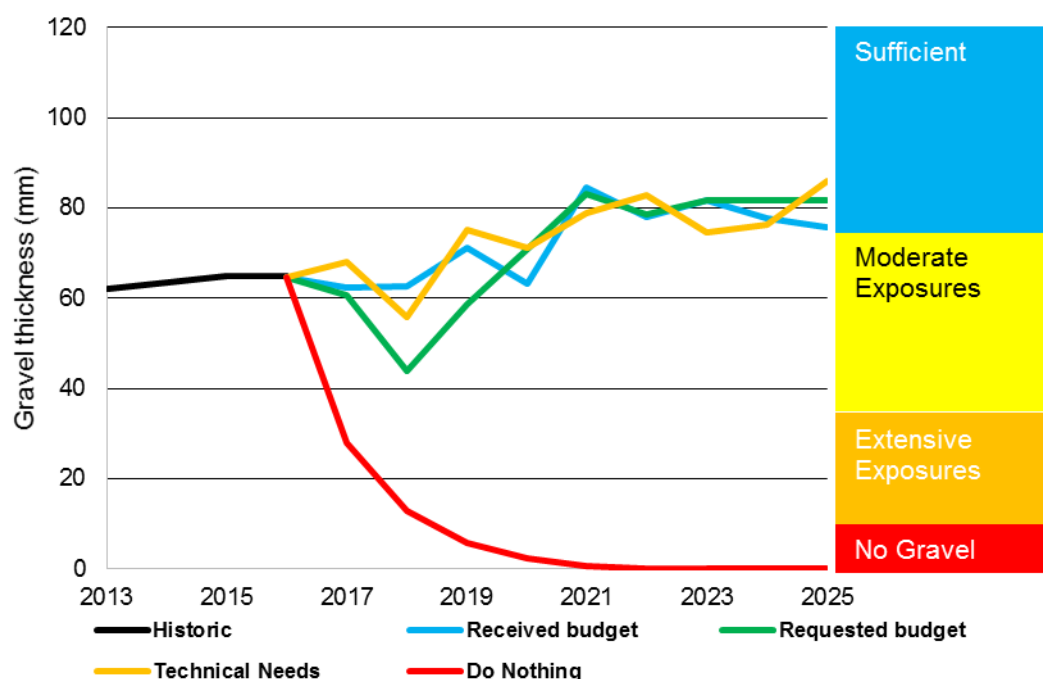


Figure 11-6: Predicted gravel thickness on unpaved roads

Figure 11-7 shows the distribution of gravel thickness for the funding scenarios. Irrespective of the funding scenario roads with inadequate wearing course material does not decrease to below 10%. A reason for this phenomenon is that as roads get upgraded to paved standards, the remaining roads in a poor condition do not necessarily increase but accounts for a smaller portion of the road network. Upgrades are done for roads with higher traffic and not necessarily roads that are in a poor condition. Roads in a poor condition might also not be of economic importance and therefore funding for the maintenance of these roads is limited.

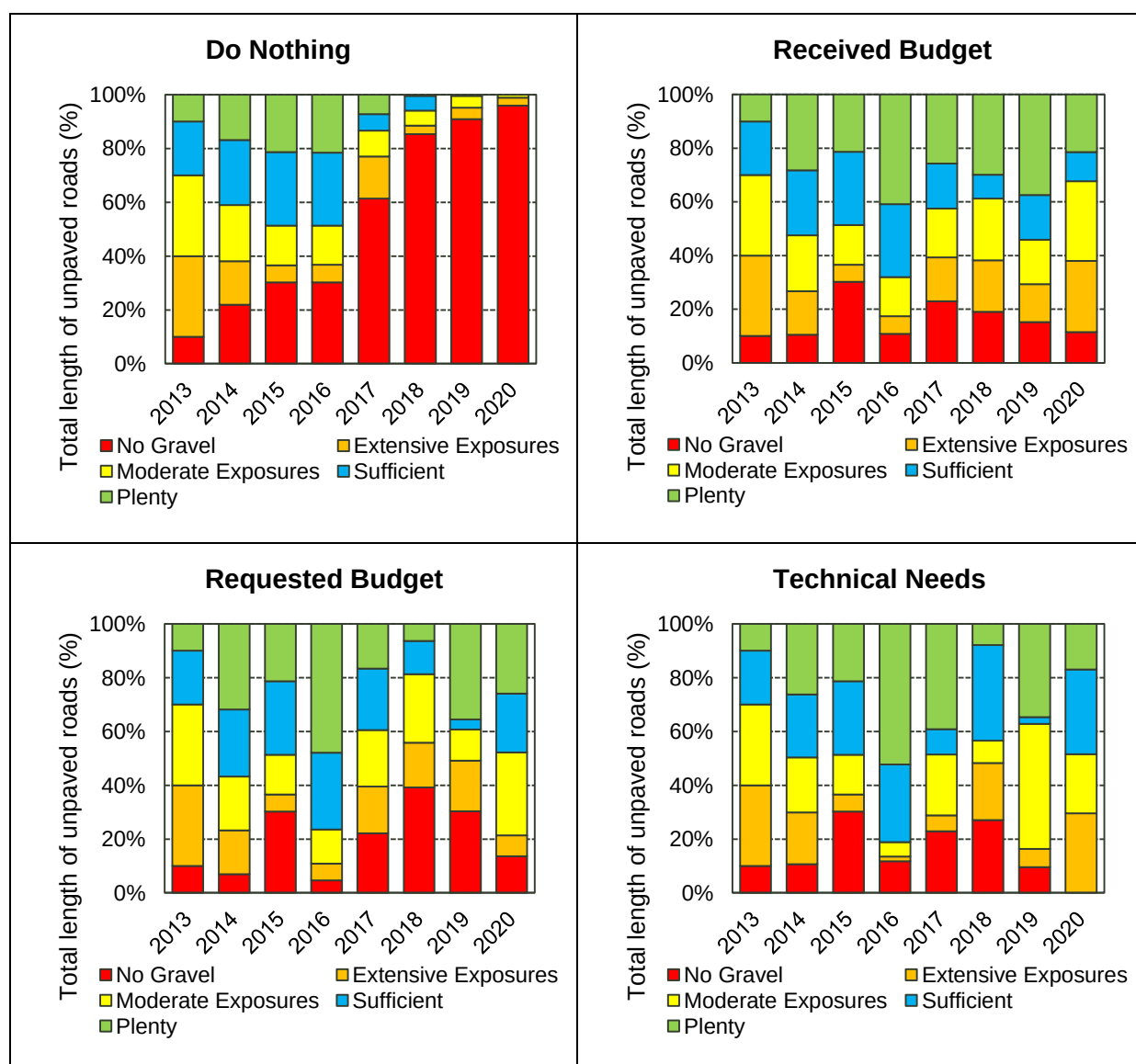


Figure 11-7: Distribution of predicted gravel thickness

11.4 Cost and Length Comparison of Received and Requested Funding

Since the condition of the network for the Received and Requested Funding Scenarios is similar for the analysis period, although the funding levels differ with US\$ 596 million, the cost and length distributions for paved and unpaved roads were assessed in detail. Figure 11-8 shows the funding level for paved roads and the affected lengths. For the Received Funding Scenario an average of US\$ 154 million annually is used to treat 2,947 km over a five year period. This is very similar to the US\$ 152 million used on the Requested Funding Scenario to treat 2,968 km of paved road in five years.

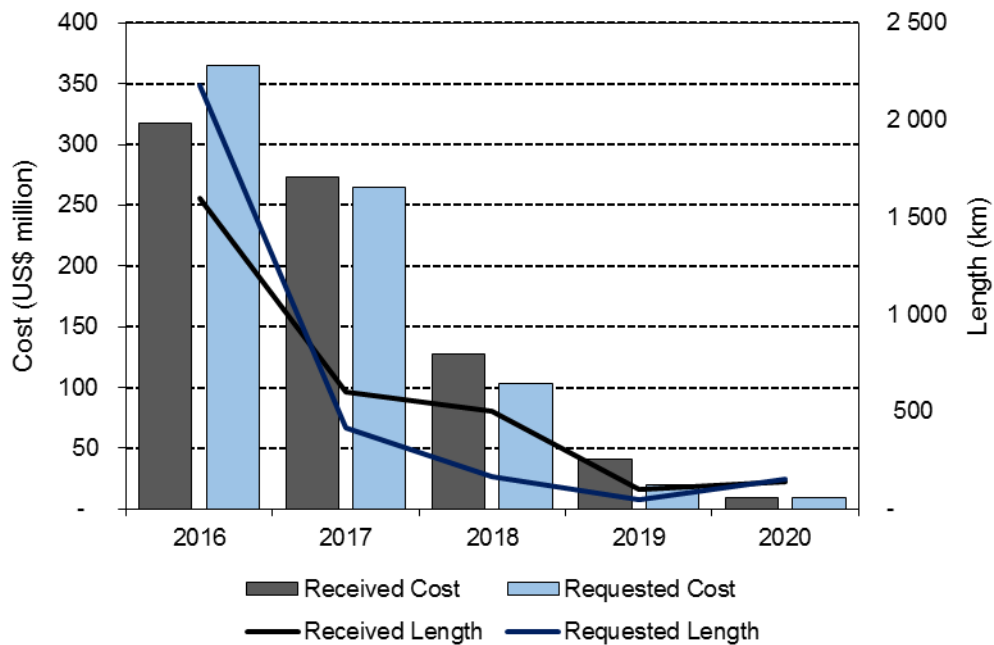


Figure 11-8: Comparison of Received and Requested Funding Scenarios and the associated length of the paved network affected

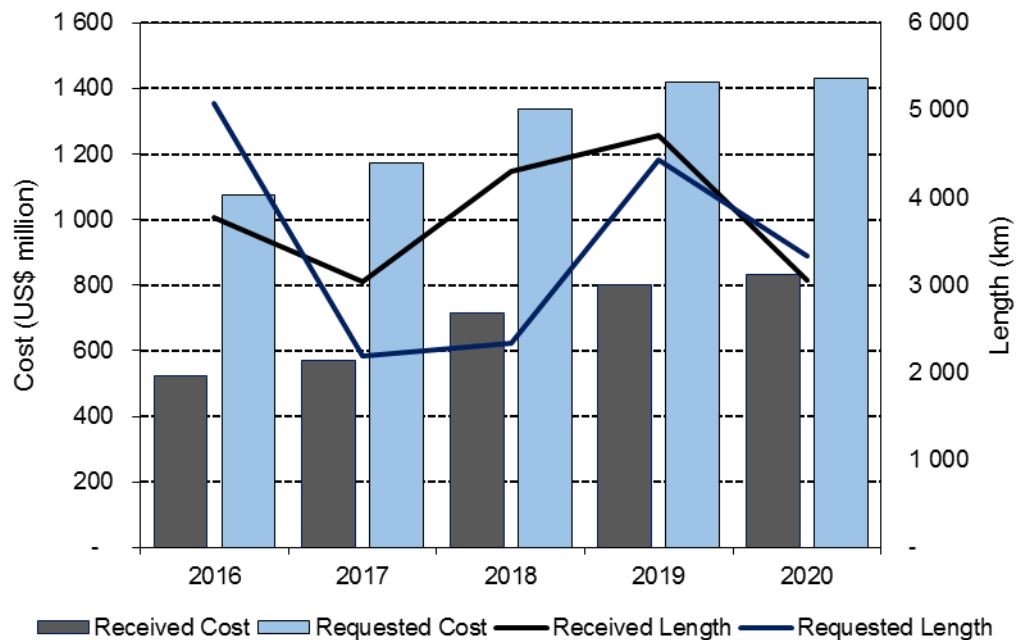


Figure 11-9: Comparison of Received and Requested Funding Scenarios and the associated length of the unpaved network affected

Figure 11-9 shows the funding levels for unpaved roads and the affected lengths. For the Received Funding Scenario an average of US\$ 689 million is used annually to treat 18,867 km over a five year period. The Requested Funding Scenario uses US\$ 1,286 million to treat 17,363 km over a five year period. The split of funds between regravelling and upgrading was assessed and is shown in Figure 11-10 and Figure 11-11.

Figure 11-10 indicates that the Received Funding Scenario allocates more funds towards regravelling than the Requested Funding Scenario. US\$ 40 million is annually allocated towards the regravelling of 13,825 km of unpaved roads over a period of five years on the Received Budget. The Requested Budget allocates US\$ 24 million annually towards the regravelling of 7,837 km of road.

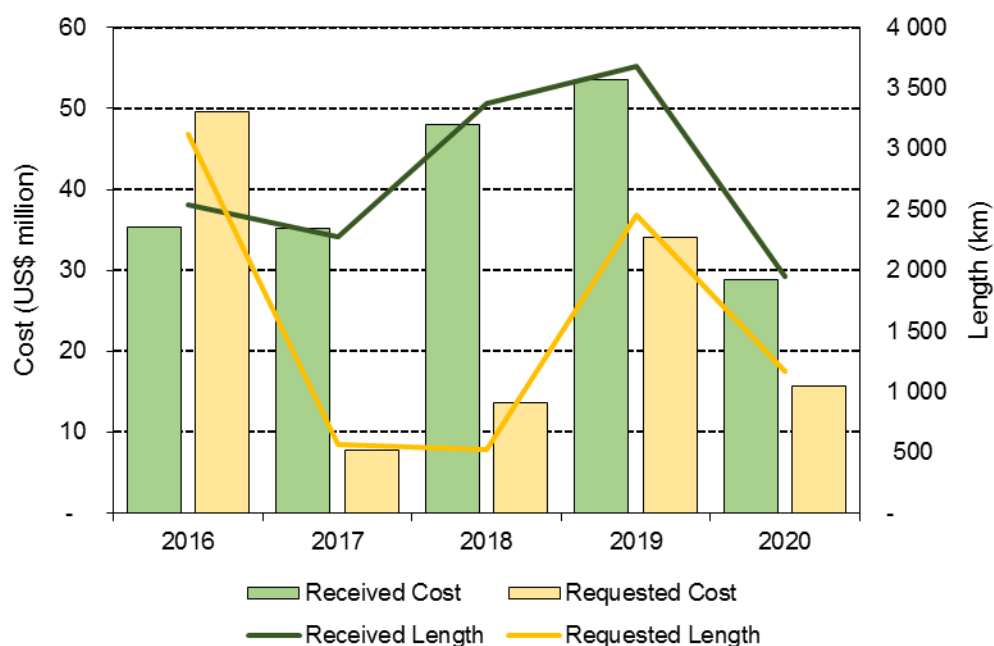


Figure 11-10: Comparison of Received and Requested Funding Scenarios and the associated length of the unpaved network regravelled

Figure 11-11 shows that the Requested Budget allocates more funds towards the upgrading of unpaved roads than the Received Budget. The Received Budget allocates US\$ 649 million annually for the upgrading of 5,042 km over a period of five years whereas the Requested Budget allocates US\$ 1,262 million annually towards the upgrading of 9,526 km over a five year period.

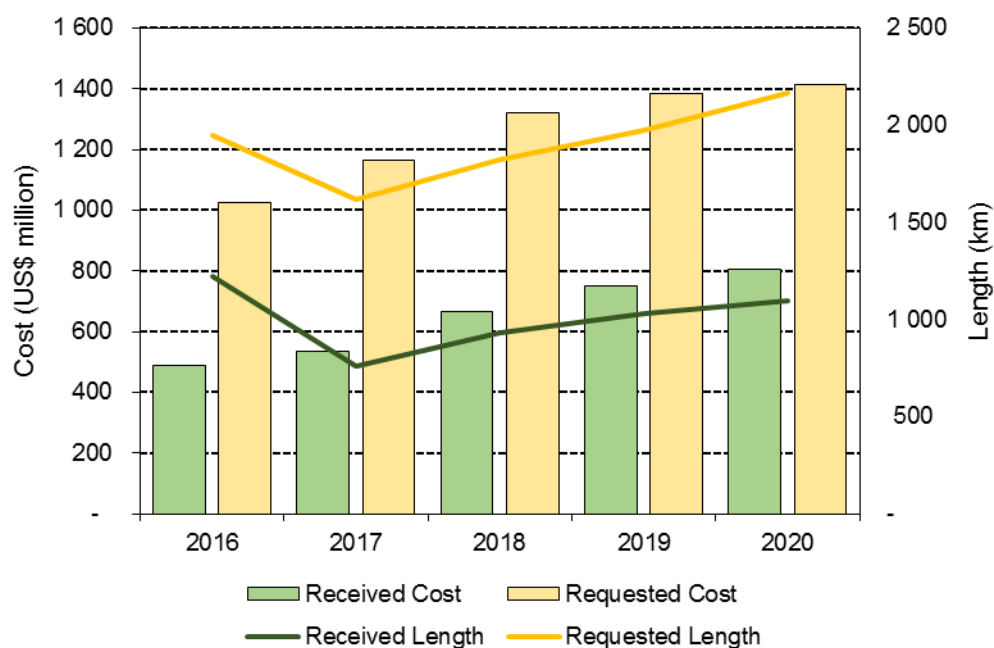


Figure 11-11: Comparison of Received and Requested Funding Scenarios and the associated length of the unpaved network upgraded

The current condition of paved roads is good therefore fund allocations lean more towards the upgrading of heavily trafficked unpaved roads. Therefore if higher funding levels are available a larger portion of the fund allocation would go towards the maintenance and upgrading of unpaved roads.

12 Vehicle Operating Costs

Summary:

- **The current VOC is US\$ 1,040 per 1000 vehicle- kilometres travelled.**
- **VOC by 2025 for:**
 - **Do Nothing scenario is US\$ 1,126**
 - **Received Budget is US\$ 800**
 - **Requested Budget is US\$ 725**

The current average VOC per thousand vehicle-kilometres driven on the UNRA road network is calculated at approximately US\$ 1,040 (or US\$1.04 per vehicle-kilometre). If nothing is done to the road network the average VOC will increase to US\$ 1,126 per 1000 vehicle-kilometres by 2025. This value could be reduced to US\$ 800 per 1000 vehicle-kilometres travelled by 2025 should the Received Budget be adopted. This value will further decrease to US\$ 725 if the Requested Budget is allocated. Figure 12-1 shows the trend for VOC on the road network.

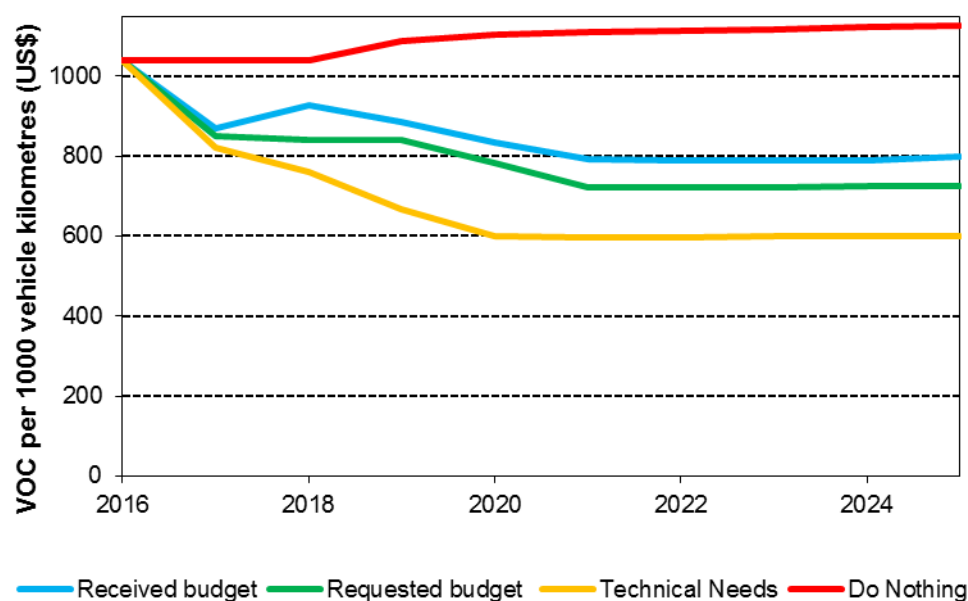


Figure 12-1: Vehicle operating cost per 1000 vehicle-kilometres travelled on the UNRA road network

13 Excess User Costs

Summary:

- **The current EUC is US\$ 483 per 1000 vehicle-kilometres travelled.**
- **The EUC by 2025 for:**
 - **Do Nothing scenario is US\$ 503**
 - **Received Budget is US\$ 407**
 - **Requested Budget is US\$ 407**

Excess User Costs are defined as the excess or additional user costs that would not have been experienced if the road had acceptable roughness. Excess user cost is calculated on roads with an average IRI of greater than 3.1 IRI. The current Excess User Cost per thousand vehicle-kilometres driven on the UNRA road network is calculated at approximately US\$ 483. If nothing is done to the road network the average EUC will increase to US\$ 503 per 1000 vehicle-kilometres by 2025. This value could be reduced to US\$ 407 per 1000 vehicle-kilometres travelled by 2025 should either the Received Budget or Requested Budget be adopted. Figure 13-1 shows the trend for EUC on the road network.

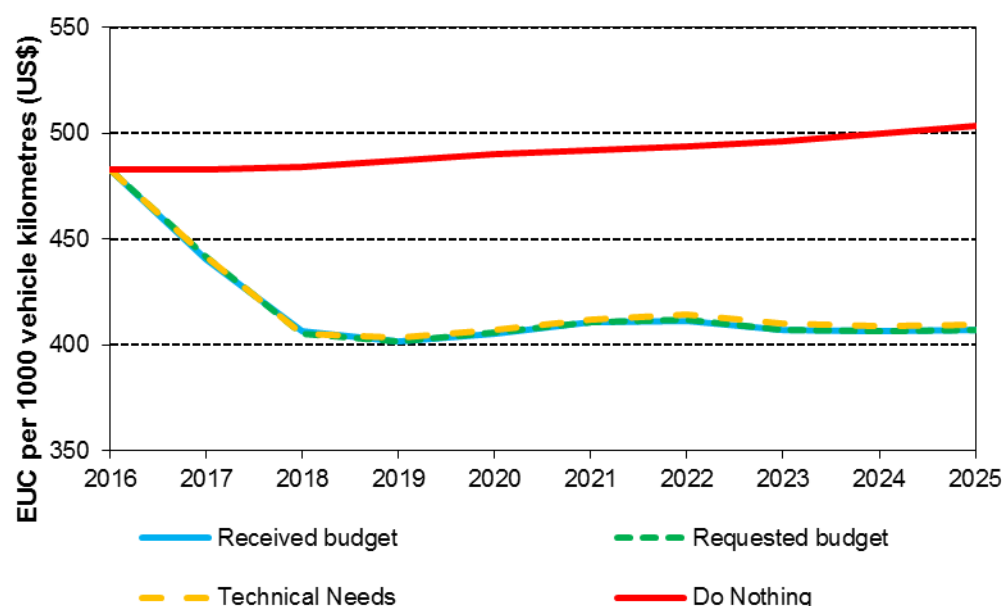


Figure 13-1: Excess user cost per 1000 vehicle-kilometres travelled on the UNRA road network

14 Asset Value

Summary:

- *The Current Depreciated Replacement Cost (CDRC) for the paved road network is US\$ 3,2 billion.*
- *The Current Replacement Cost (CRC) for the UNRA paved network is US\$ 4 billion.*
- *For the unpaved network the CDRC is US\$ 84 million and the CRC is US\$ 194 million*

Asset value is a measure of the worth of a road network and the value of a pavement consists of the following costs:

- The ground preparation work,
- the road foundations and
- the road structural layers.

Over time, pavement structures deteriorate and the value of the structural layers decreases accordingly. Eventually, reconstruction (i.e. heavy rehabilitation or regravelling) becomes necessary and if the deterioration of the pavement layers continues even further a stage will be reached where the pavement has no functionality and the value of the structural layers is zero.

The value of any road is calculated as the value it would have if it was new, minus the cost of taking it from its present condition to the “very good” or new condition. The amount subtracted represents the cost of eliminating any deficiency the road may have. This concept is similar to the generally accepted accounting practice of calculating an asset’s “book value”, which equates to cost minus accumulated depreciation. Asset value is expressed as the Current Depreciated Replacement Cost (CDRC), i.e. in terms of current US\$ values. The potential or maximum asset value refers to the replacement cost in 2016 US\$ values, or Current Replacement Cost (CRC).

Note that the cost of road furniture such as signs, bridges and culverts are not included in this calculation of asset value.

It is estimated that the Current Replacement Cost, CRC of the paved road network is approximately US\$ 4,052 million (US\$ 4,0 billion); thus assuming all roads are in a very good condition. The asset value associated with the current condition of the paved road network, i.e. the Current Depreciated Replacement Cost (CDRC), was calculated at US\$ 3,201 million (US\$ 3,2 billion). The CDRC is currently 79 per cent of the CRC, or alternatively stated there is asset loss of 21%. A CDRC to CRC ratio greater than 80% is indicative of a well-maintained road network and the current performance of the paved road network is therefore fine. Figure 14-1 shows the predicted CRC and CDRC for the paved portion of the UNRA network.

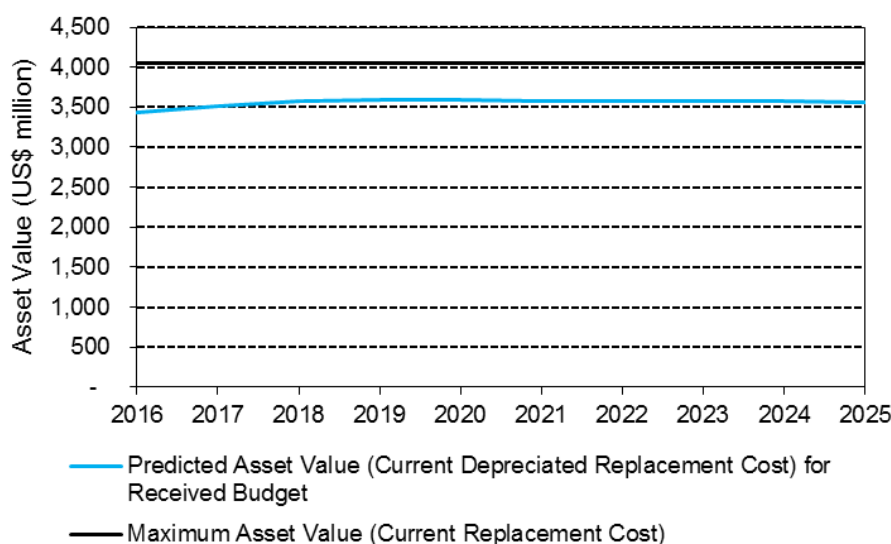


Figure 14-1: Predicted Asset Value for the Received Budget versus the Maximum Asset Value for Paved Roads

It is estimated that the Current Replacement Cost, CRC of the unpaved road network is approximately US\$ 194 million; thus assuming all gravel roads have plenty of gravel material. The asset value associated with the current gravel thickness of the unpaved road network, i.e. the Current Depreciated Replacement Cost (CDRC), is approximately US\$ 84 million. The CDRC is currently 43 per cent of the CRC.

It can be noted that the Maximum Asset value and the CDRC of the unpaved roads increase over the analysis period. This is due to current unpaved roads that are upgraded to paved standards during the analysis.

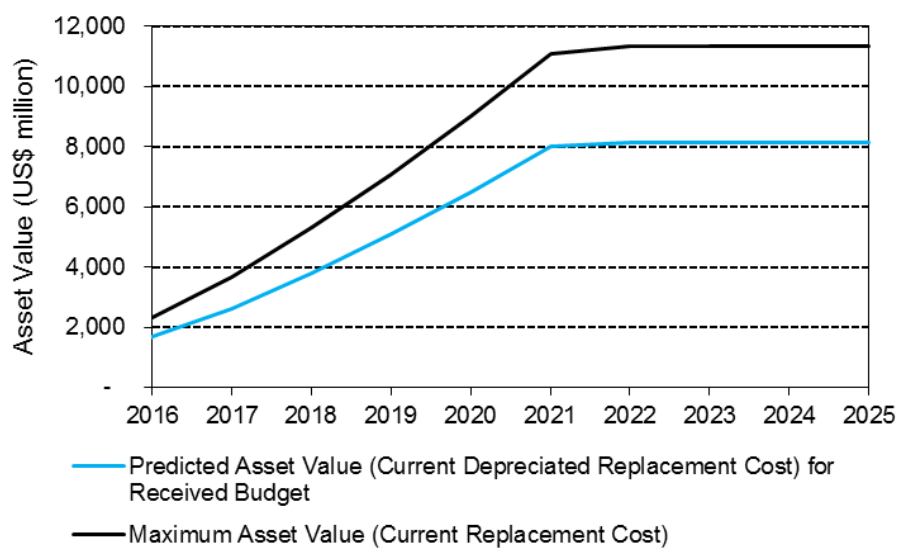


Figure 14-2: Predicted Asset Value for the Received Budget versus the Maximum Asset Value for Unpaved Roads

15 Summary

This report concludes with a summary of the most important findings of the strategic analysis.

This analysis has highlighted interesting aspects of the UNRA road network. These are summarised in bullet points below for each of the paved and unpaved road networks.

Paved roads

- The paved road network comprises 4,157 km, which is only 20% of the total UNRA road network.
- 69% of the paved network was visually assessed and was analysed in this strategic analysis.
- The analysed network carries 38% of all traffic, and 6,697 million vehicle-kilometres are travelled annually on this network.
- One quarter of the network carries more than 10,000 vehicles per day, and half carries less than 5,000 vehicles per day.
- The overall average VCI of the paved network is 78%, which indicates a network in “good” condition. Compared to the previous analysis, there has been an improvement in the overall condition of 5%.
- The percentage roads in poor and very poor condition (the backlog) is 2%, which is 8% less than the World Bank recommended maximum of 10% for this category. The backlog for rehabilitation is therefore insignificant.
- The percentage of roads in fair condition is 23%, which is 2% less than the experienced based recommended maximum of 25%. This shows that good preventive maintenance strategies are being followed.
- A Technical Needs analysis of reseal, light and heavy rehabilitation needs, given current road conditions and covering 5 years from 2016, identified needs to the value of US\$ 375 million on 1,893 km of road. This is only US\$ 75 million per annum.
- Under the Received Funding Scenario, the optimisation allocated on average US\$ 154 million per annum to paved roads over a five year period. This represents 18% of the total

available funding of this funding scenario, allowing 1,536 km road to be rehabilitated, and 1,411 km road to be resealed over the five years.

- The result of the allocation of the Received Funding Scenario is that the current paved road condition will remain “good”, and the expected length in poor and very poor condition will be 61 km after 5 years. Excess User Costs will be reduced to US\$ 407 per 1000 vehicle-kilometres which is a 16% decrease achieved for road users compared to the current EUC.
- Maintenance of the full paved network should be strived for given its important function. This is indeed achievable under the Received Funding Scenario. The optimisation analysis indicated that under higher funding scenarios, such as the Requested Funding Scenario, the optimum allocation to paved roads is similar to the optimum allocation of the Received Funding Scenario.

Unpaved roads

- The unpaved network of UNRA is extensive and amounts to 16,388 km of road. This represents 80% of the total network.
- 82% of the unpaved network was visually assessed and was analysed in this strategic analysis.
- The analysed network carries 62% of all traffic, and 11,125 million vehicle-kilometres are travelled annually on this unpaved network.
- 95% of the network carries more than 500 vehicles per day. 85% of unpaved road travel concerns volumes between 1,000 and 5,000 vehicles per day. This implies a high need for upgrading to paved standards.
- The average VCI of the unpaved network is 61%, which indicates a “fair” condition. Compared to the previous analysis, the condition has remained the same.
- The percentage of roads in poor and very poor condition is 25% which is 15% higher than the World Bank recommended maximum of 10% for this category. The backlog for regravelling is therefore significant.
- The average gravel thickness of the unpaved network is 65 mm.
- A Technical Needs analysis of investment requirements for regavel and upgrading to paved standards, given current road conditions and traffic volumes, and covering 5 years

from 2016, identified investment requirements to the value of US\$ 5,569 million on 9,197 km of road.

- The regravelling need, based on the Technical Needs determination, is US\$ 2 million, or 510 km approximately over the 5 year period.
- The upgrading from unpaved to paved road investment need, based on the Technical Needs determination, is US\$ 5,567 million, or 8,687 km approximately over the 5 year period.
- Under the Received Funding Scenario, the optimisation allocated on average US\$ 689 million per annum to unpaved roads over a five year period. This represents 82% of the total available funding of this funding scenario, allowing 5,042 km road to be upgraded to paved standards, and 13,825 km road to be regravelled over the five years.

The optimisation analysis indicated that under higher funding scenarios than the Received Funding Scenario, such as the Requested Funding Scenario, the optimum allocation to paved roads is similar to the optimum allocation of the Received Funding Scenario, but more funds are available for the high need for upgrading of unpaved roads to paved standards.

Table 15-1 compares the optimum funding allocations of Received Budget with the Requested Budget scenario.

Table 15-1: Funding allocation of the Requested Budget Compared to Received Budget (US\$ million and km summed for the first 5 years)		
	Requested Budget	Received Budget
Rehabilitation allocation	US\$ 680 (1,536 km) US\$ 136 per annum	US\$ 690 (1,536 km) US\$ 138 per annum
Reseal allocation	US\$ 85 (1,432 km) US\$ 17 per annum	US\$ 80 (1,411 km) US\$ 16 per annum
Regravelling allocation	US\$ 120 (7,867 km) US\$ 24 per annum	US\$ 201 (13,825 km) US\$ 40 per annum
Upgrade to paved roads allocation	US\$ 6,310 (9,526 km) US\$ 1,262 per annum	US\$ 3,246 (5,042 km) US\$ 649 per annum
TOTAL PER ANNUM	US\$ 1,439	US\$ 843

Appendix A – Committed Projects

Appendix B – M&R's